



DATA WAREHOUSING AND BUSINESS INTELLIGENCE

Assignment 2

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Step 1: Data Source for Assignment 2

1.1 Data Source Overview & Justification

The primary data source utilized for this Business Intelligence and OLAP implementation is the Logistics_DWH Data Warehouse. This repository was successfully designed, developed, populated via an SSIS ETL pipeline, and validated during Assignment 1.

Operating as the central analytical foundation for a synthetic Class 8 trucking company, this data warehouse transforms highly normalized, transactional operational data (OLTP) into a denormalized, read-optimized analytical structure (OLAP). The dataset (sourced originally from the *Logistics-Operations-Database*) provides a rich, multi-year history of fleet operations. It contains sufficient volume and complexity spanning multiple subject areas like freight movement, asset maintenance, and fuel consumption making it the ideal candidate for advanced multidimensional cube modeling and Power BI reporting.

Data Set Link: [logistics-operations-database](#)

1.2 Schema Design & Architecture

To support complex, multi-process analytics, the data warehouse is structured using a **Fact Constellation Schema (Galaxy Schema)**. This architecture allows multiple distinct business processes to be analyzed together by sharing a common set of conformed dimensions.

Unlike a traditional Star Schema that centers on a single business process, a Fact Constellation architecture accommodates multiple distinct, yet related, business processes by sharing a common set of conformed dimensions. This allows end-users to analyze completely different metrics (e.g., comparing truck maintenance costs against freight revenue) within the same BI dashboard.

The data warehouse consists of the following components:

Fact Tables (Measuring Business Events):

- **FactTrips:** The core operational fact table, modeled as an *accumulating snapshot*. The grain is defined as exactly one row per completed logistics trip. It tracks the complete lifecycle of a freight movement, containing additive measures like Total Revenue and Distance, alongside accumulating metrics such as Transaction Creation Time, Completion Time, and Total Processing Hours.
- **FactFuelPurchases:** A transactional fact table designed to track fleet expenses and fuel efficiency. The grain is one row per individual fuel transaction. Key measures include Fuel Gallons Purchased and Total Transaction Cost, allowing for cost-per-mile analytics when cross-referenced with trip data.

- **FactDeliveryEvents:** A transactional fact table tracking warehouse and facility performance. The grain is one row per facility delivery or pickup event. It captures critical supply chain metrics, specifically measuring Detention Minutes and boolean On-Time Status flags.
- **FactMaintenance:** A transactional fact table monitoring equipment health and repair expenditures. The grain is one row per service ticket. It contains financial measures (Maintenance Costs) and operational measures (Downtime Hours) to analyze asset reliability.

Conformed Dimensions (Providing Business Context): The schema utilizes 7 shared conformed dimensions that link the fact tables together, providing the descriptive attributes necessary for filtering, slicing, and dicing the data in SSAS and Excel:

- **DimDate:** The standard time dimension essential for temporal analytics. It contains attributes to support the required hierarchical drill-downs (e.g., Year, Quarter, Month, Day).
- **DimCustomer:** Contains client-specific descriptive attributes, allowing revenue and performance to be grouped by specific client hierarchies or broader Industry Types.
- **DimDriver:** Contains demographic and professional attributes, such as Driver Name, License Class, and Experience Level, enabling performance tracking at the human-resource level.
- **DimTruck:** Implemented as a Slowly Changing Dimension (SCD Type 2). This is a critical architectural feature that uses ValidFrom, ValidTo, and IsCurrent flags to accurately preserve historical changes in fleet equipment status, ensuring historical reporting remains accurate even if a truck changes status or configuration.
- **DimTrailer:** Categorizes the hauled assets, containing attributes like Equipment Type (e.g., Dry Van, Refrigerated) and Capacity.
- **DimFacility:** Contains geographic locations, physical addresses, and map coordinates for warehouses and docks, enabling spatial analytics and location-based performance tracking.
- **DimRoute:** Contains standard origin-to-destination pairings and typical baseline distances, allowing the business to identify their most profitable or heavily trafficked freight lanes.

The screenshot shows the 'Data Source Designer' window with the 'General' tab selected. The 'Data source name' is 'Logistics_DW'. The 'Provider' field is empty. The 'Connection string' is 'Provider=MSOLEDBSQL19.1;Data Source=.\SQLDEV;Integr'. There is an 'Edit...' button next to the connection string. Below this is a section for 'Data source references' with a checkbox 'Maintain a reference to another object in the solution' which is unchecked. Below the checkbox is a dropdown menu with the text 'Create a data source based on an existing data source'. Below the dropdown is an empty text box. Below the 'Data source references' section are three fields: 'Isolation' set to 'ReadCommitted', 'Query timeout (in seconds)' set to '0', and 'Maximum number of connections' set to '10'. At the bottom is a 'Data source description' text box. At the very bottom of the window are three buttons: 'OK', 'Cancel', and 'Help'.

Data Source Designer

General Impersonation Information

Data source name: Logistics_DW

Provider:

Connection string: Provider=MSOLEDBSQL19.1;Data Source=.\SQLDEV;Integr Edit...

Data source references

☐ Maintain a reference to another object in the solution

Create a data source based on an existing data source

Isolation: ReadCommitted

Query timeout (in seconds): 0

Maximum number of connections: 10

Data source description:

OK Cancel Help

Figure 1 - Configuring the Data Source connection in Analysis Services to link to the underlying Logistics_DW SQL Server database.

1.3 Data Warehouse Entity-Relationship (ER) Diagram

The following Entity-Relationship diagram illustrates the physical dimensional model of the Logistics_DWH. As shown, the four centralized fact tables are interconnected via the surrounding conformed dimensions. This exact physical structure serves as the underlying Data Source View (DSV) for the SSAS Multidimensional Cube developed in Step 2.

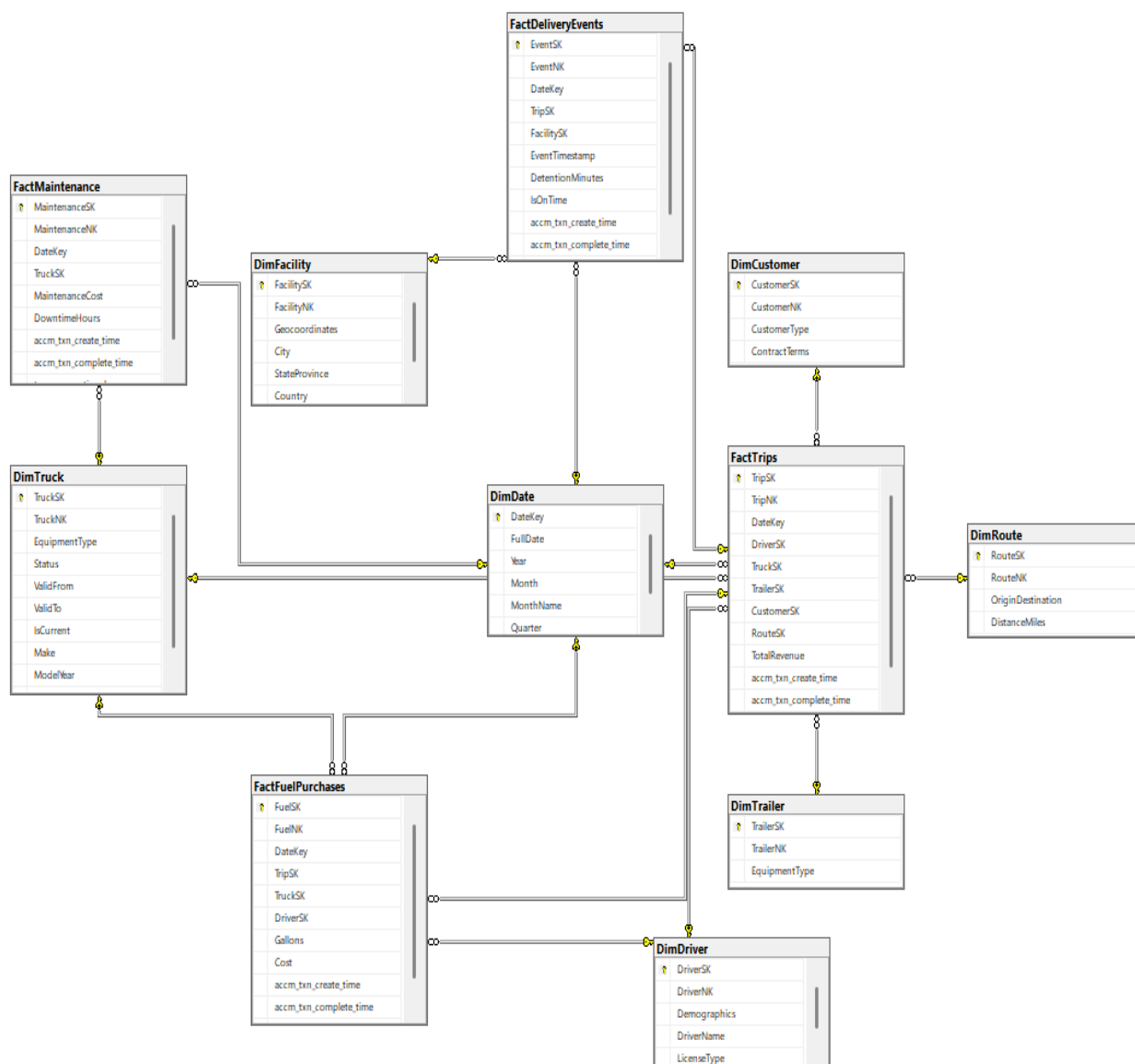


Figure 2 – Logistics Data Warehouse – Galaxy Schema

Step 2: SSAS Cube Implementation

2.1 Project Initialization & Data Source Configuration

To transition the data from a relational storage model to a multidimensional analytical model, a new **Analysis Services Multidimensional and Data Mining Project** was created using Visual Studio (SQL Server Data Tools).

The first step in the implementation was establishing the connection to the underlying data. A new **Data Source** was configured to connect directly to the Logistics_DWH database hosted on the SQL Server Developer Edition instance. Windows Authentication/Service Account credentials were used to ensure secure data access during cube processing.

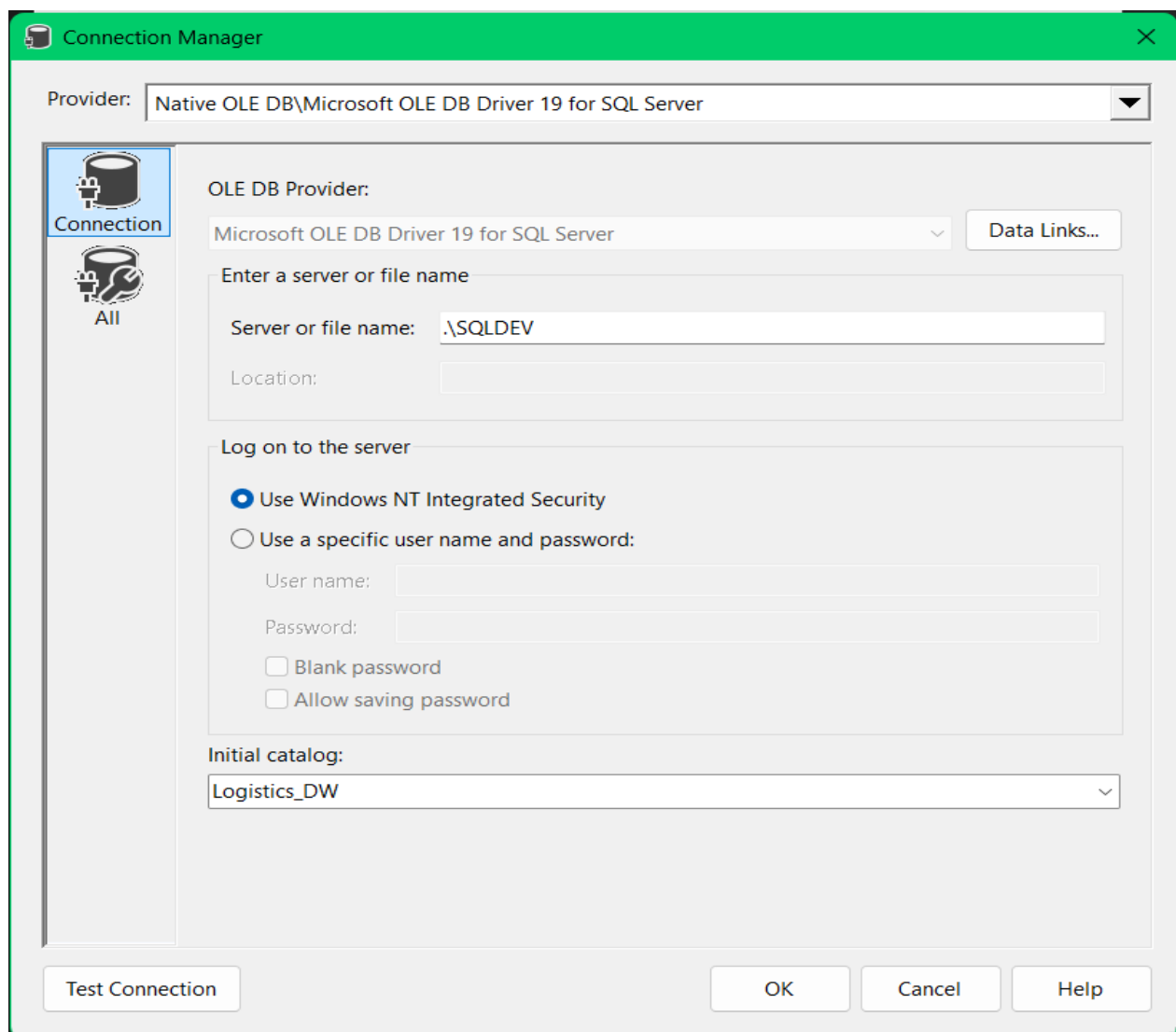


Figure 3 - Configuring the OLE DB Connection Manager using Windows NT Integrated Security to establish a secure link to the Logistics_DW database on the local .\SQLDEV server.

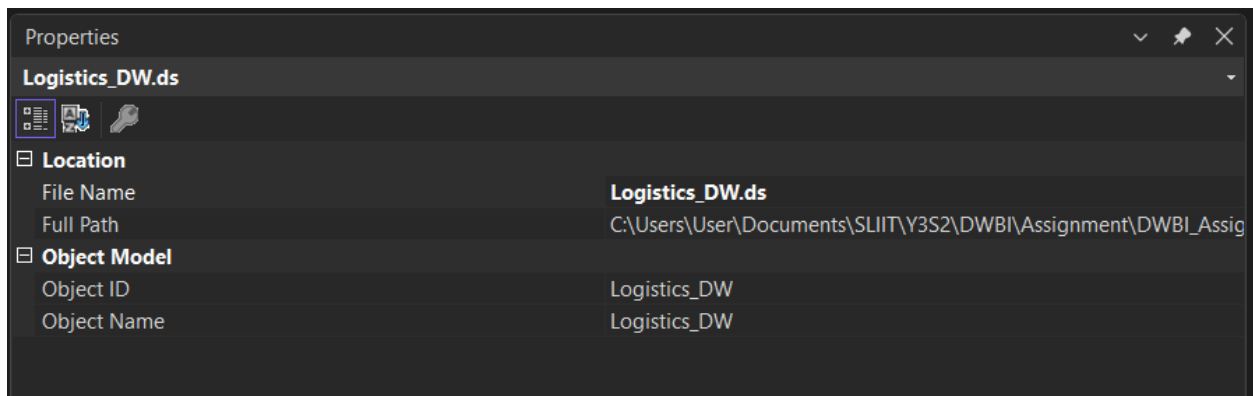


Figure 4 - Verifying the properties and file path of the finalized Logistics_DW.ds Data Source object within the Analysis Services project solution.

2.2 Designing the Data Source View (DSV)

Once the connection was established, a **Data Source View (DSV)** was created. The DSV acts as a logical abstraction layer between the physical database and the SSAS cube.

All four fact tables (FactTrips, FactFuelPurchases, FactDeliveryEvents, FactMaintenance) and the seven conformed dimensions (DimDate, DimDriver, DimTruck, etc.) were imported into the DSV workspace. The primary key to foreign key relationships established during the database creation were automatically recognized, preserving the Fact Constellation schema structure for the multidimensional model.

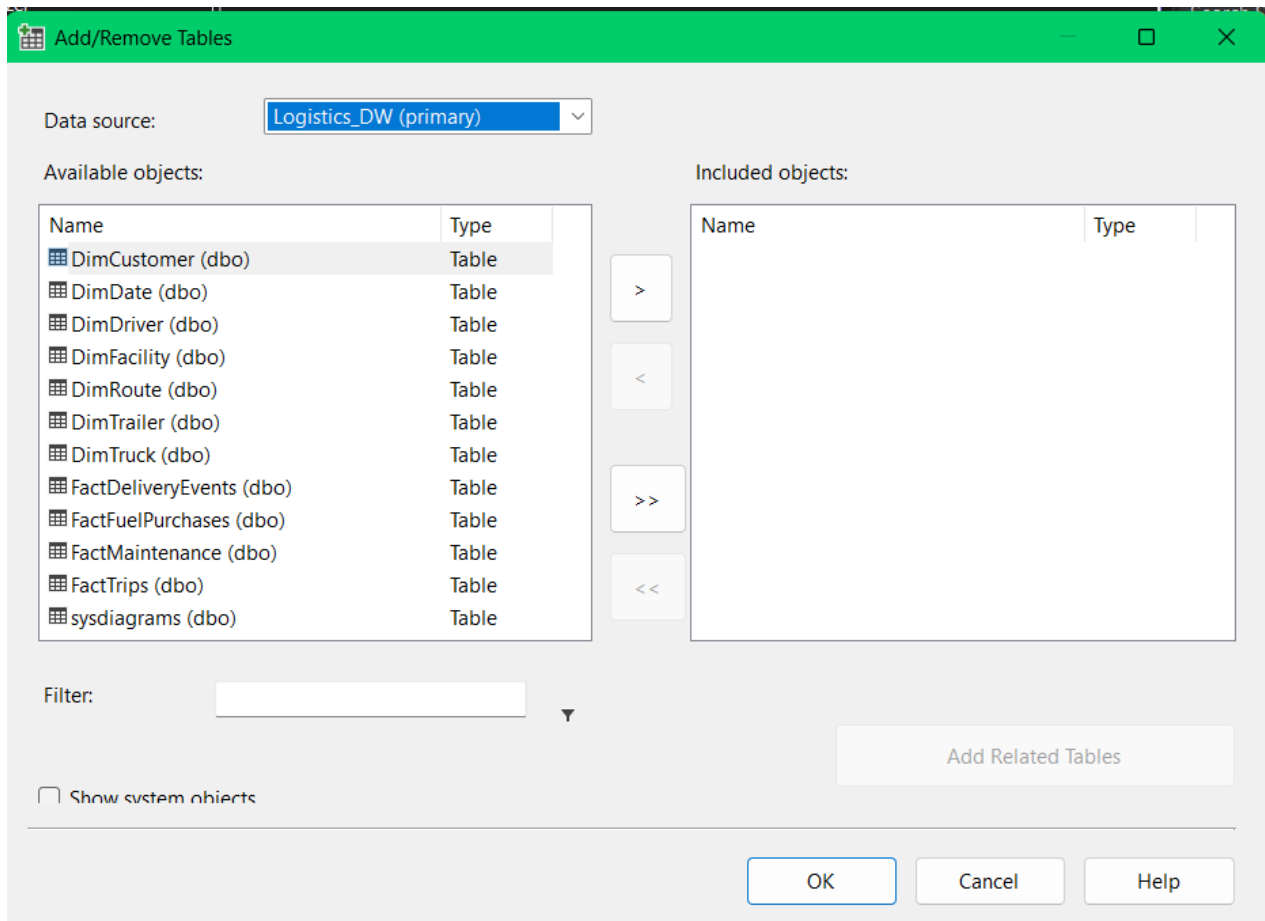


Figure 5 - Initializing the Data Source View (DSV) wizard by reviewing the available tables from the underlying Logistics_DW data source.

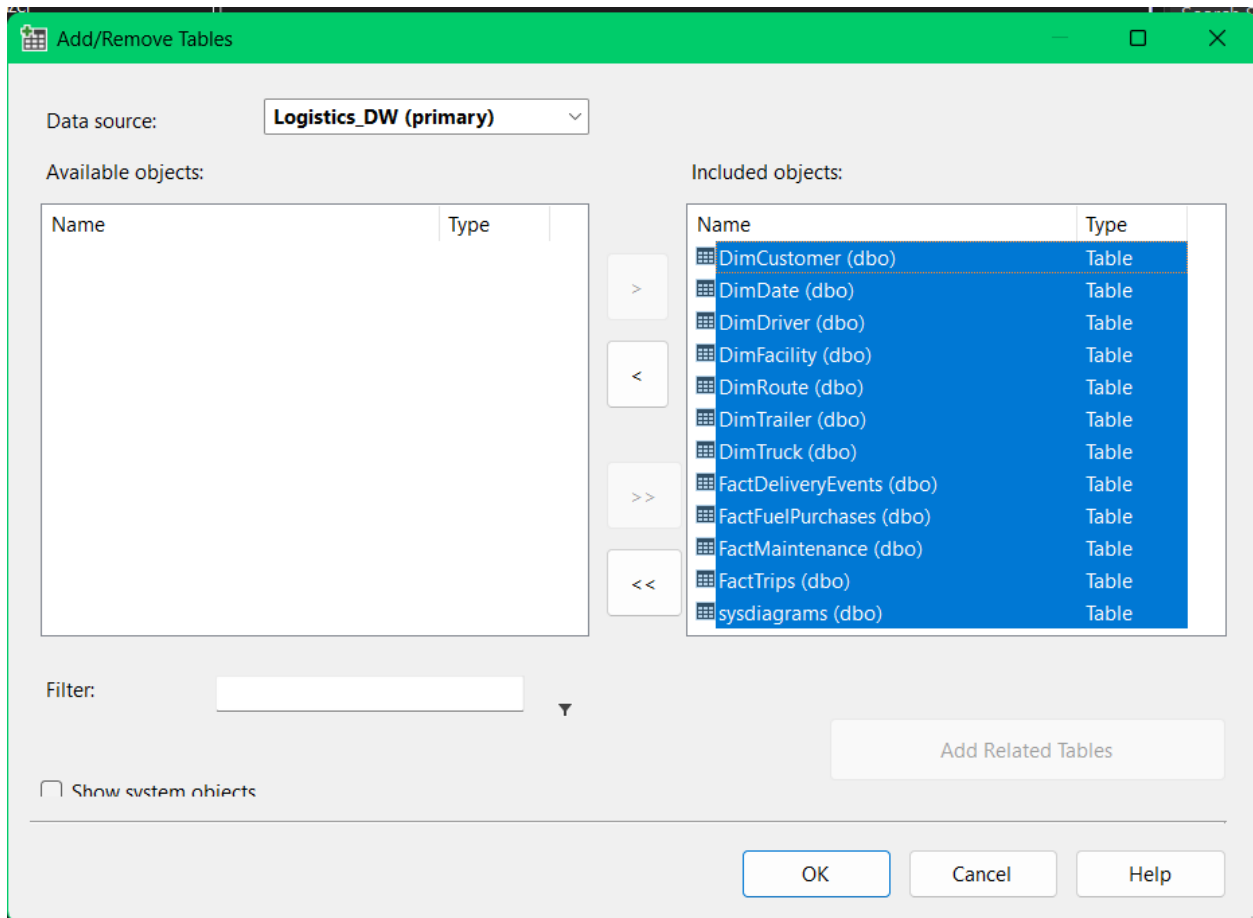


Figure 6 - Selecting and transferring all necessary Fact and Dimension tables into the "Included objects" pane to build the data model.

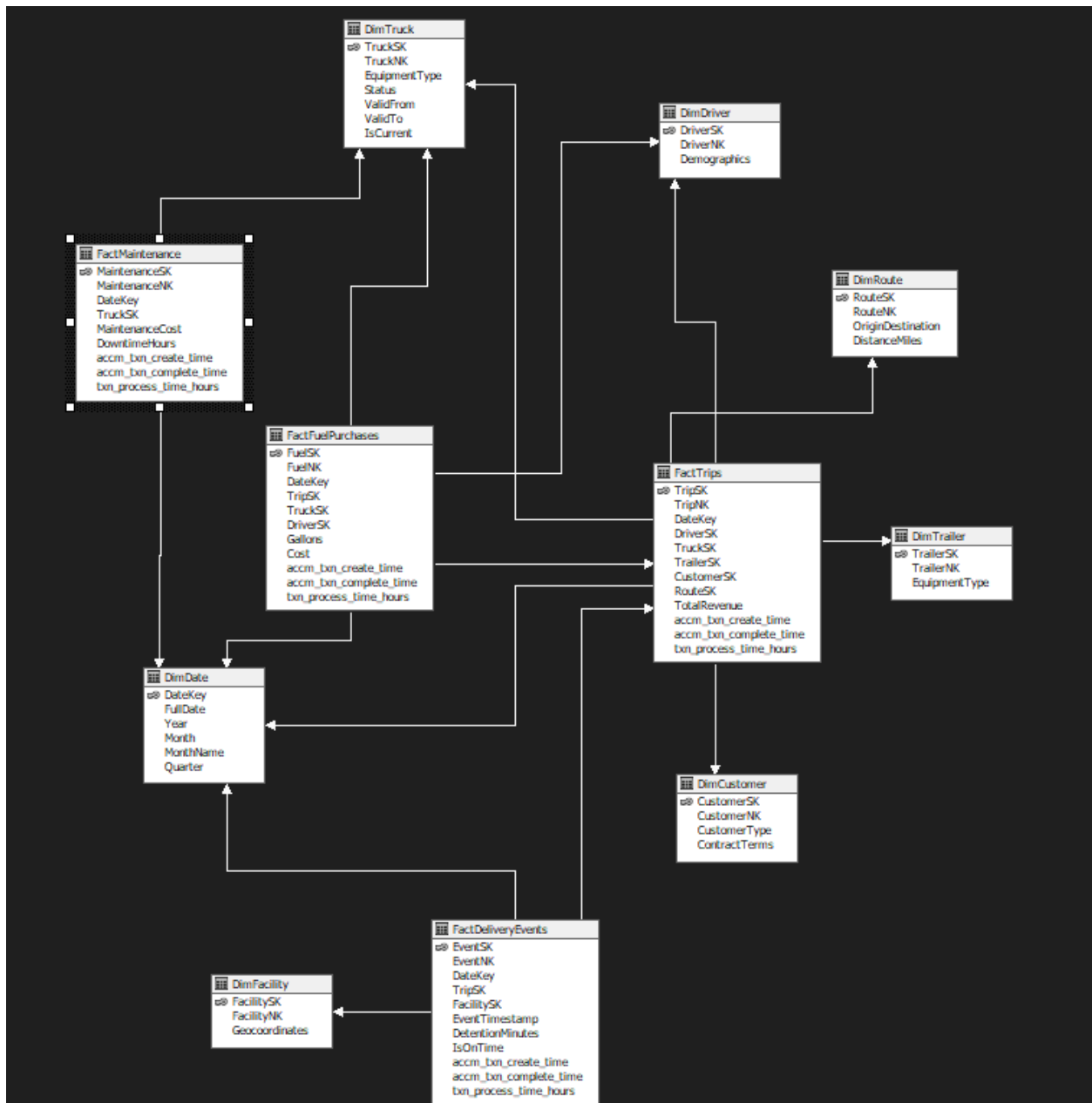
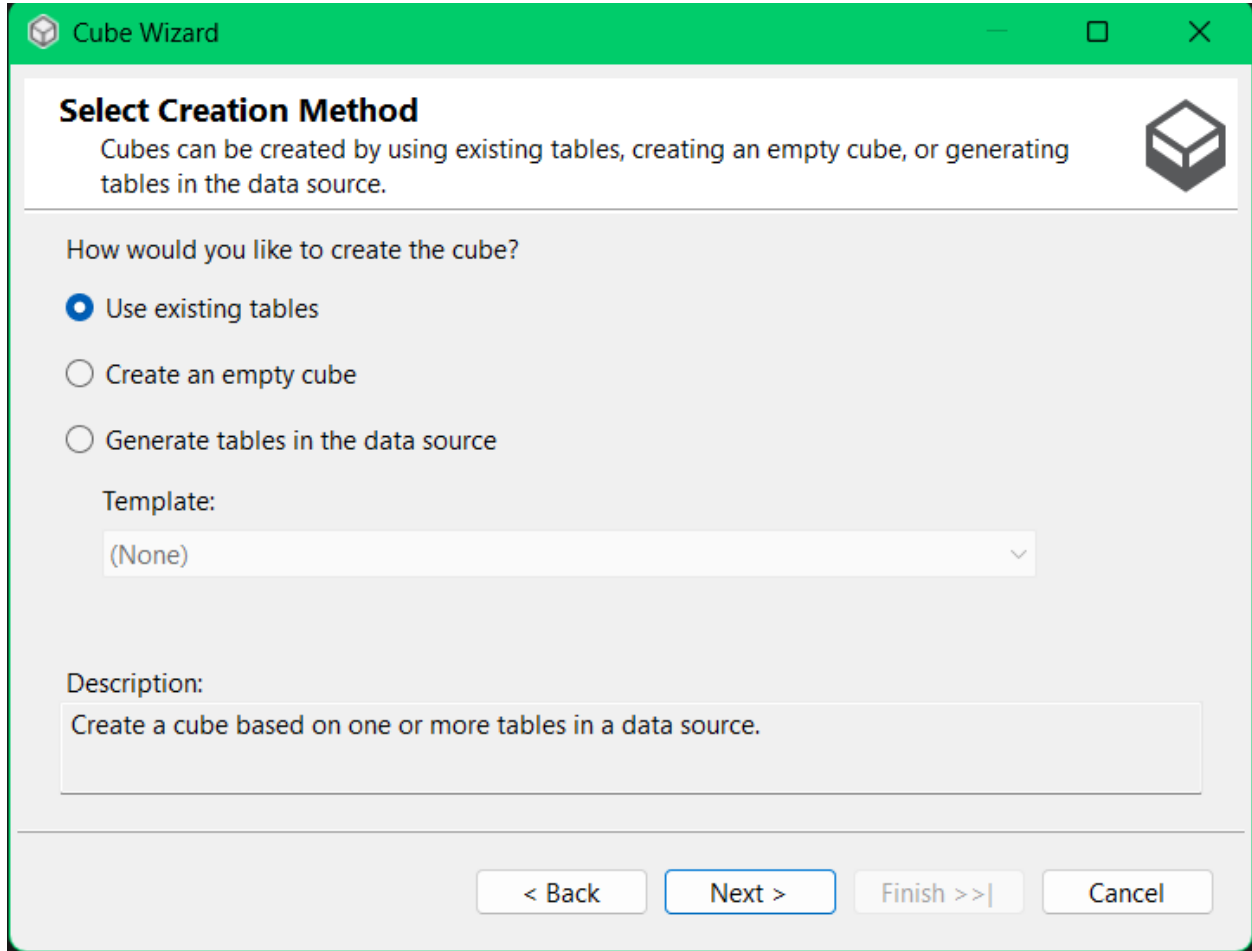


Figure 7 - The finalized Data Source View displaying the complex schema layout and established physical relationships between the four Fact tables and their respective Dimensions.

2.3 Cube Generation and Measure Definition

The core OLAP Cube was generated using the Cube Wizard based on the tables present in the DSV.



The image shows a screenshot of the 'Cube Wizard' application window. The title bar is green and contains the text 'Cube Wizard' along with standard window control icons. The main content area has a light gray background. At the top, there is a section titled 'Select Creation Method' with a subtitle: 'Cubes can be created by using existing tables, creating an empty cube, or generating tables in the data source.' To the right of this text is a small 3D cube icon. Below the subtitle, the question 'How would you like to create the cube?' is displayed. There are three radio button options: 'Use existing tables' (which is selected), 'Create an empty cube', and 'Generate tables in the data source'. Below these options is a 'Template:' label followed by a dropdown menu currently showing '(None)'. Further down is a 'Description:' label followed by a text box containing the text 'Create a cube based on one or more tables in a data source.' At the bottom of the window, there are four buttons: '< Back', 'Next >' (which is highlighted with a blue border), 'Finish >>|', and 'Cancel'.

Figure 8 - Initializing the Cube Wizard and opting to build the cube using the existing tables defined in the Data Source View.

1. **Measure Groups:** The four fact tables were explicitly selected as Measure Group tables.

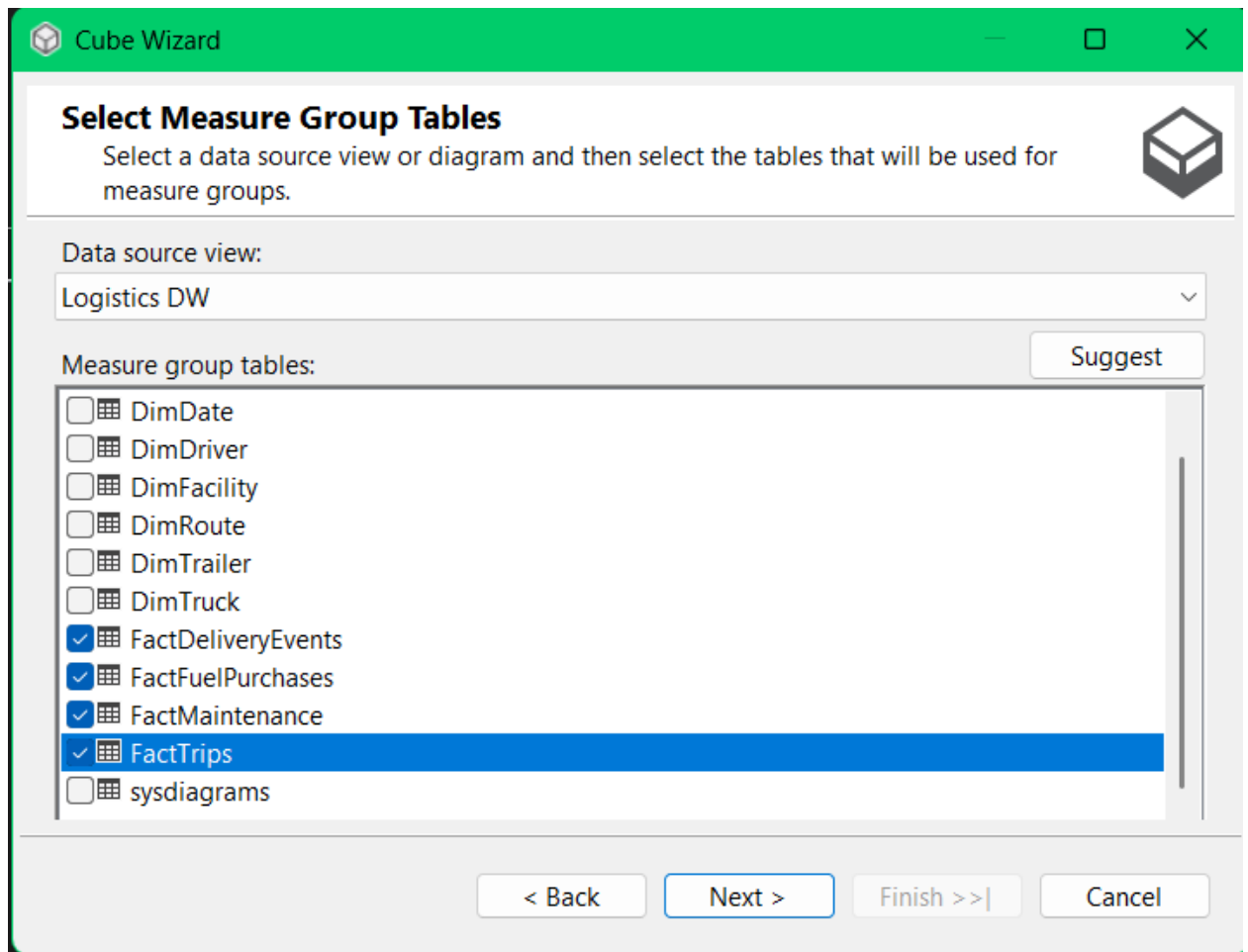


Figure 9 - Selecting the four core Fact tables (FactDeliveryEvents, FactFuelPurchases, FactMaintenance, FactTrips) to serve as the Measure Groups for the cube.

2. **Measures:** The wizard aggregated the numerical columns into quantifiable business metrics. For example:
- **Fact Trips:** Total Revenue (Sum), Distance Miles (Sum), Txn Process Time Hours (Sum).
 - **Fact Fuel Purchases:** Gallons (Sum), Cost (Sum).
 - **Fact Maintenance:** Maintenance Cost (Sum), Downtime Hours (Sum).
 - Fact Count measures were also generated for each group to allow for simple transaction volume analysis.

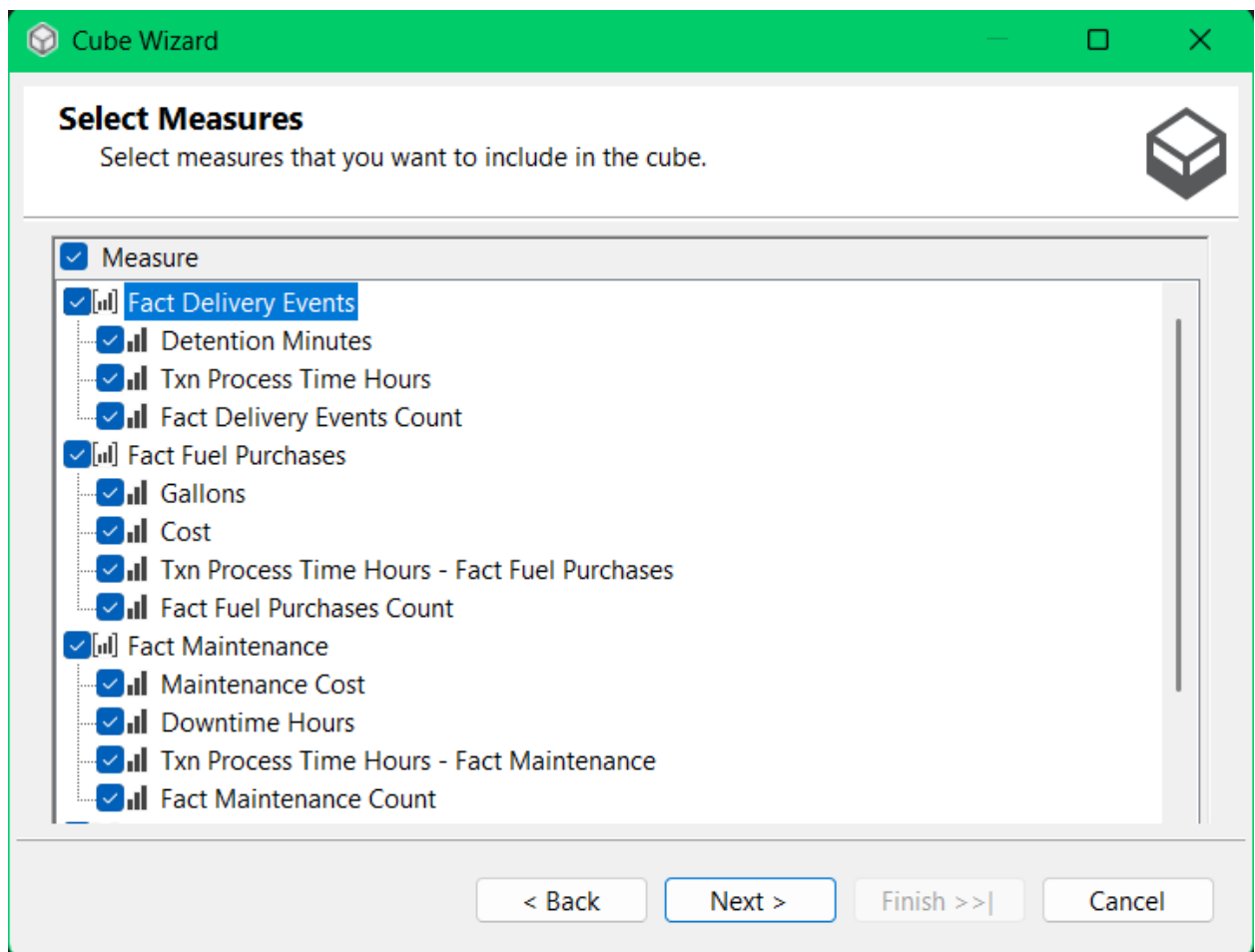


Figure 10 - Reviewing and selecting the specific numerical measures (e.g., Event Counts, Process Times, Costs) to be aggregated within each Measure Group.

3. **Dimensions:** The remaining tables were mapped as standard Cube Dimensions, linked to the measure groups via their respective surrogate keys.

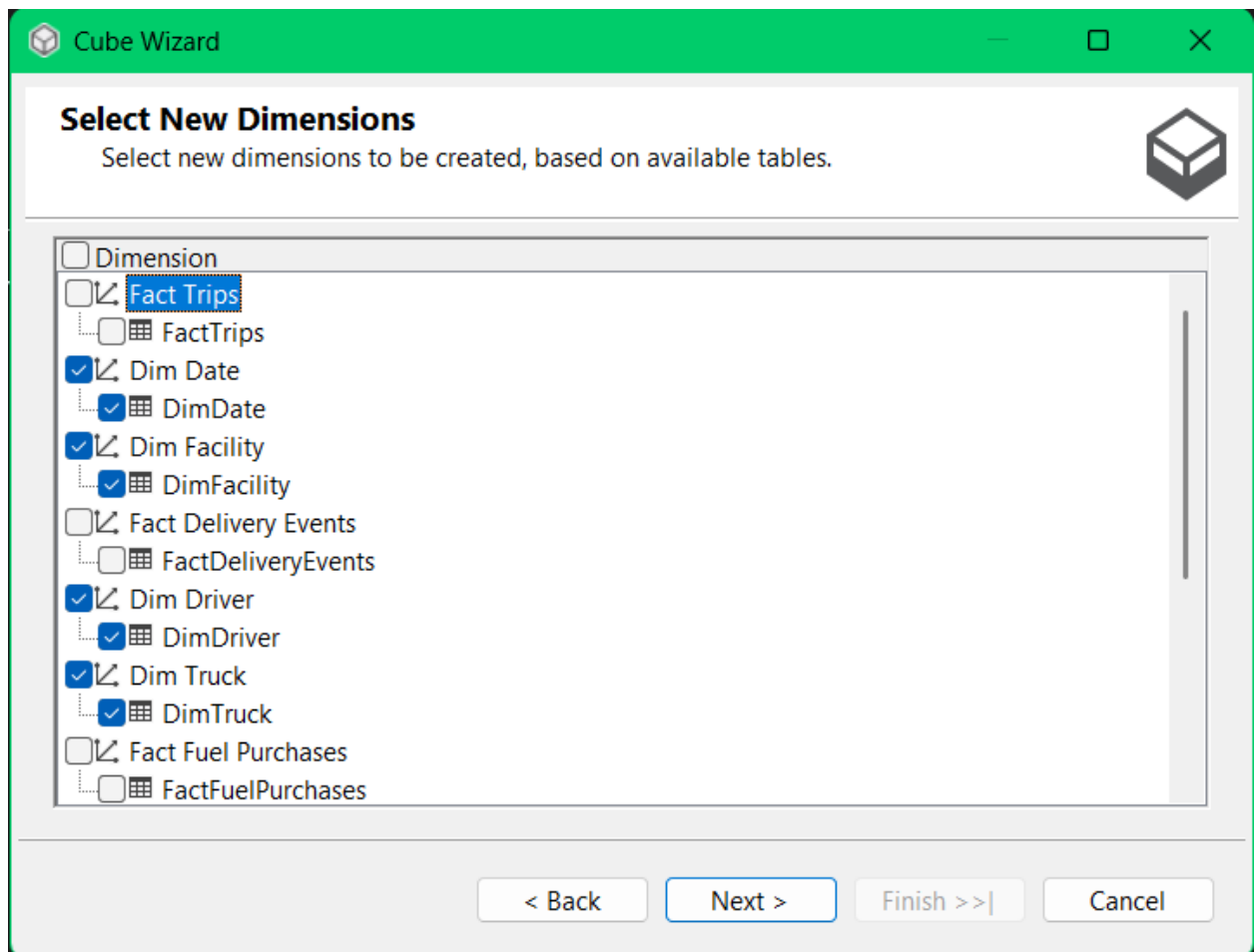


Figure 11 - Identifying and selecting the associated Dimension tables that will provide context, filtering, and grouping capabilities for the cube data.

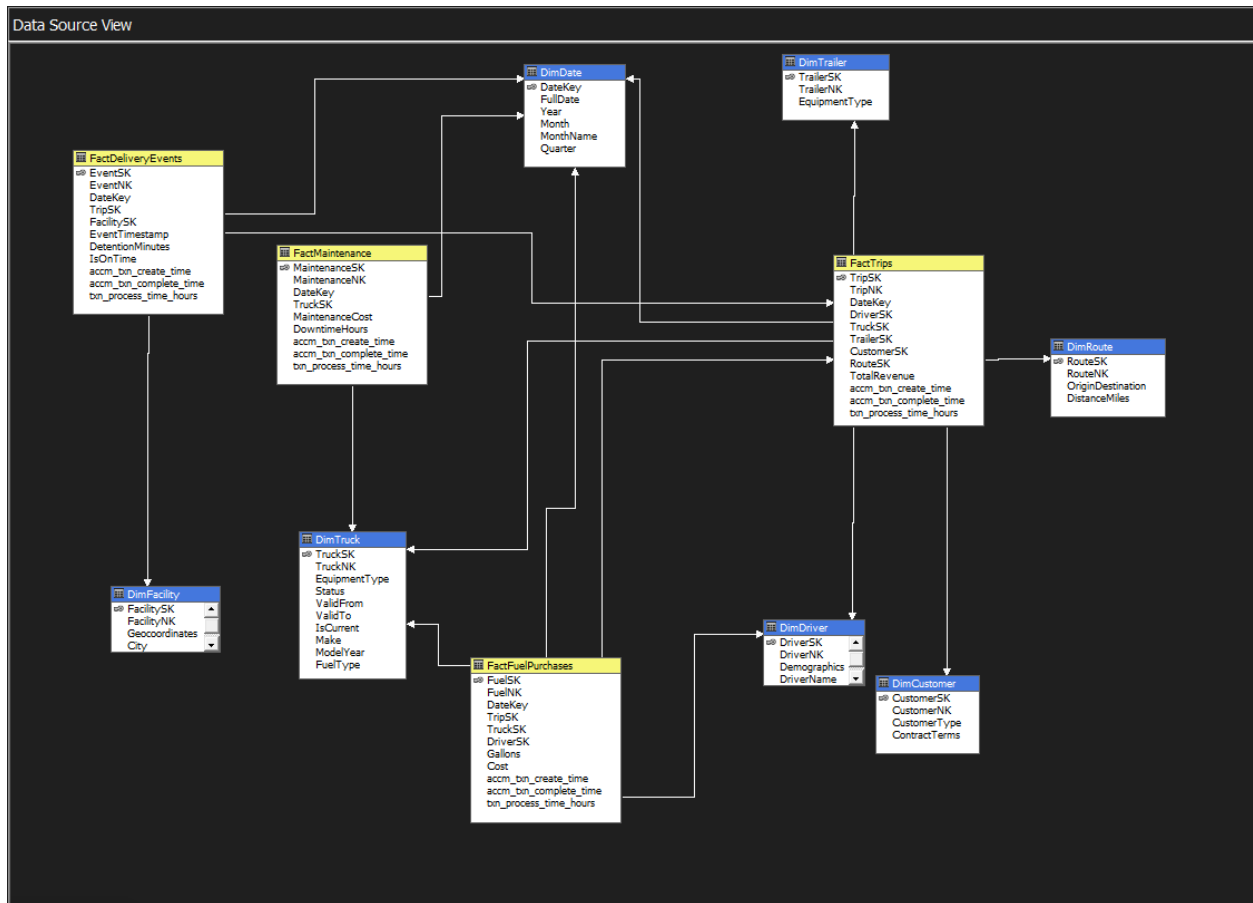


Figure 12 - Reviewing the comprehensive relational model generated for the cube, highlighting the interconnected Fact and Dimension tables.

2.4 Dimension Modeling & Hierarchy Implementation

To satisfy the requirement for hierarchical data exploration (and to prepare for the analytical drill-down requirements in Step 4), logical user-defined hierarchies were implemented across three key dimensions: Time, Geography, and Truck.

By accessing the Dimension Designer for each respective .dim file, relevant attributes were dragged from the Data Source View into the Attributes pane. To enable seamless drill-down and roll-up capabilities, hierarchies were constructed in the central Hierarchies pane. The levels for each hierarchy were organized sequentially from the highest level of aggregation to the lowest granularity.

The following hierarchies were successfully implemented:

- **Calendar Time Hierarchy (DimDate):**
 - Level 1: Year
 - Level 2: Quarter
 - Level 3: Month
 - Level 4: Full Date

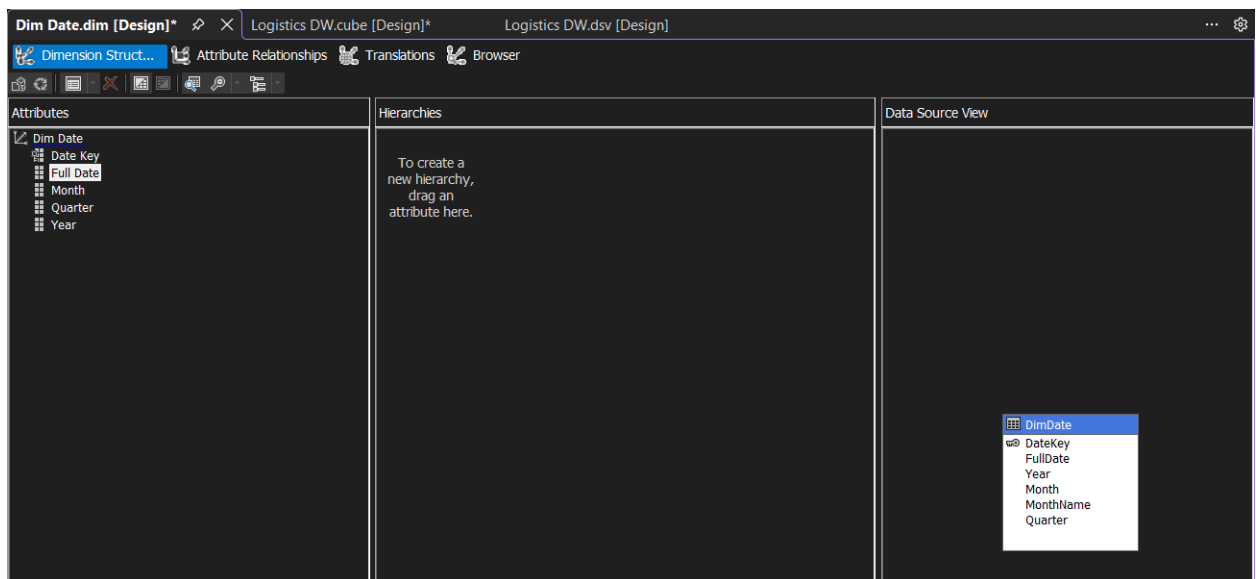


Figure 13 - Opening the Dimension Designer for Dim Date to prepare for attribute and hierarchy configuration.

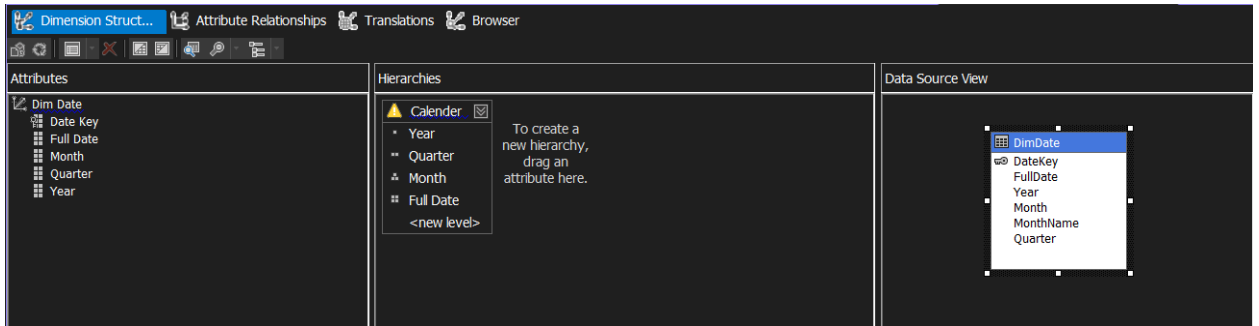


Figure 14 - Establishing the Calendar hierarchy (Year > Quarter > Month > Full Date) to enable native time-based drill-down analysis in reporting tools.

- **Geography Hierarchy (DimFacility):**
 - Level 1: Country
 - Level 2: State Province
 - Level 3: City
 - Level 4: Facility NK (Natural Key)

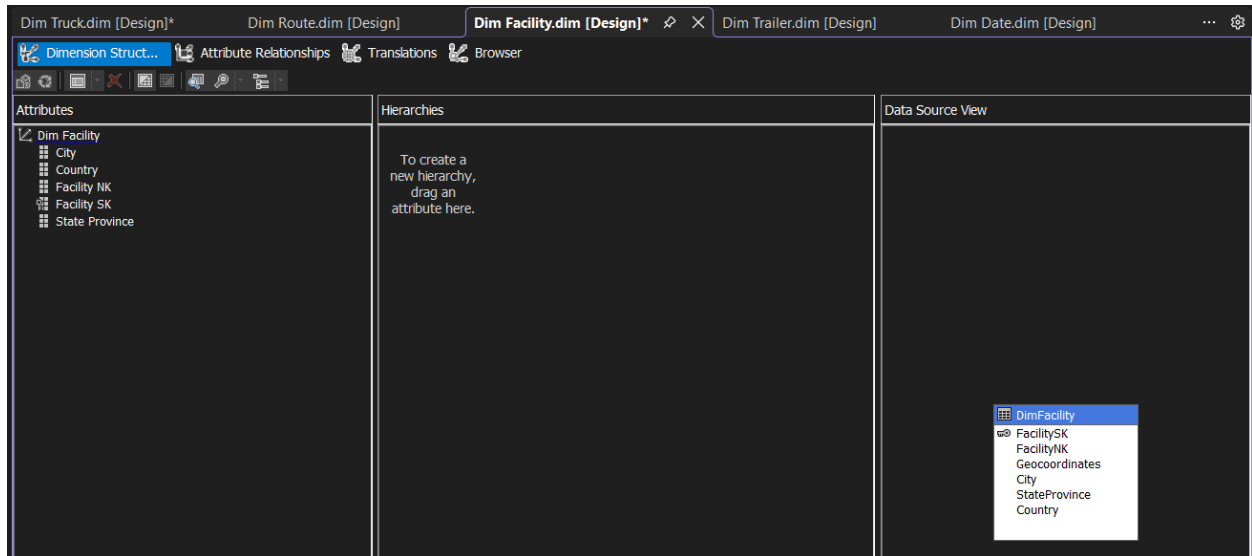


Figure 15 - Importing the relevant geographical attributes into the Dim Facility dimension structure.

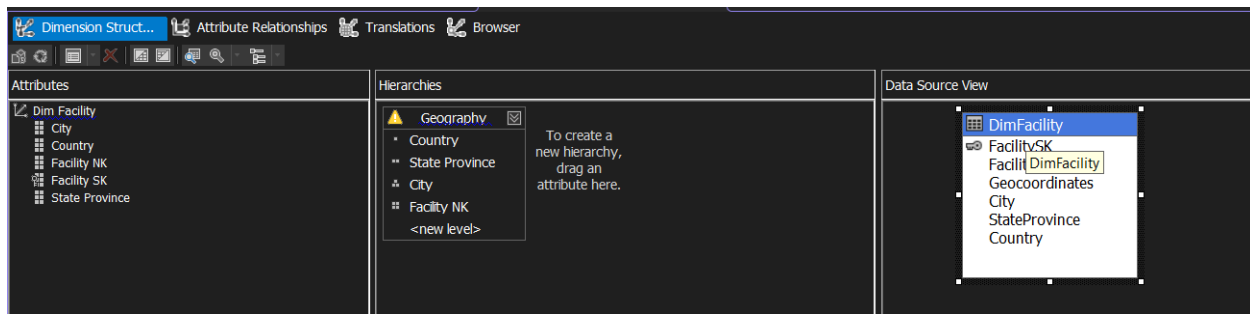


Figure 16 - Creating the Geography hierarchy (Country > State Province > City > Facility NK) to support spatial roll-ups and drill-downs.

- **Truck Hierarchy (DimTruck):**

- Level 1: Make
- Level 2: Model Year
- Level 3: Truck NK (Natural Key)

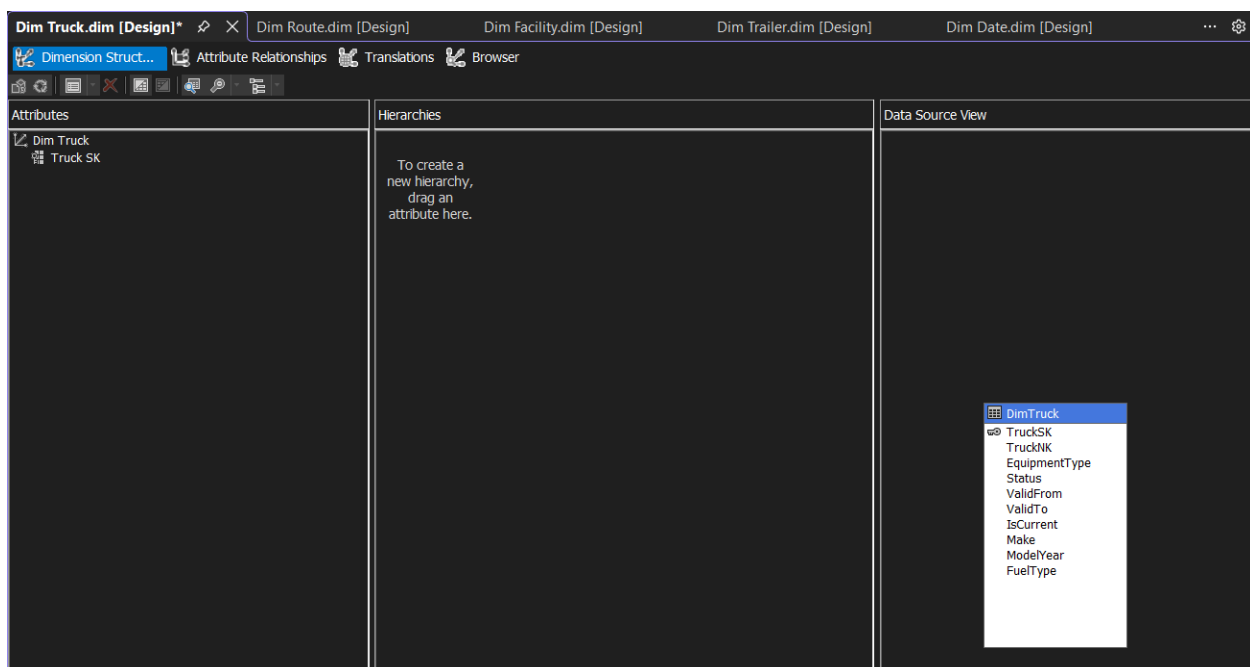


Figure 17 - Reviewing the established attributes for the Dim Truck dimension within the Dimension Designer interface.

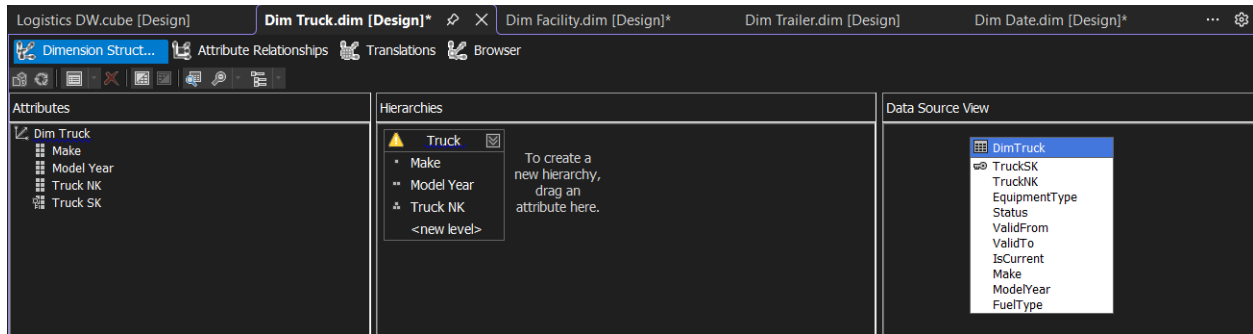


Figure 18 - Creating the Truck hierarchy (Make > Model Year > Truck NK) within the Dim Truck dimension to enable vehicle categorization drill-downs.

Finally, attribute relationships were established within the dimension structures to ensure accurate aggregations and optimized query performance when users navigate through the various levels of the data cube.

2.5 Cube Deployment and Processing

To make the multidimensional cube accessible for client-side analytical tools (such as Excel and Power BI), the project required deployment to the SQL Server Analysis Services (SSAS) engine, followed by a full data processing execution.

- **Deployment Configuration:** The project properties were configured to target the local SSAS multidimensional server instance (.\\SQLDEV). The deployment process compiled the underlying XMLA scripts and successfully generated the empty cube schema, measure groups, and dimensional structures on the target server.

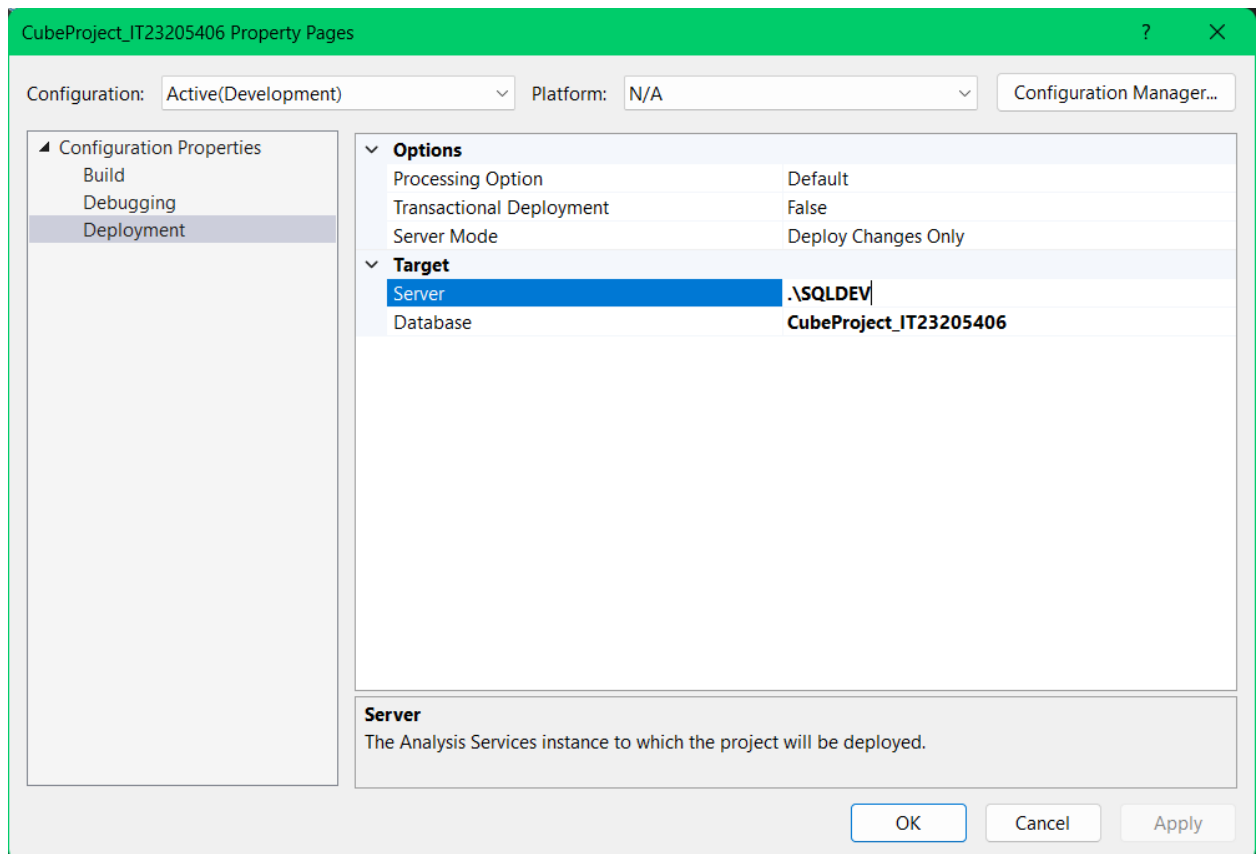


Figure 19 - Configuring the project deployment properties to target the local .\\SQLDEV Analysis Services instance.

- **Data Processing:** Following deployment, a "Process Full" command was executed on the deployed cube. This critical step queried the underlying SQL Server relational Data Warehouse, extracted the transactional data points, calculated the aggregations according to the predefined measure definitions, and physically populated the OLAP cube structures and dimension members.

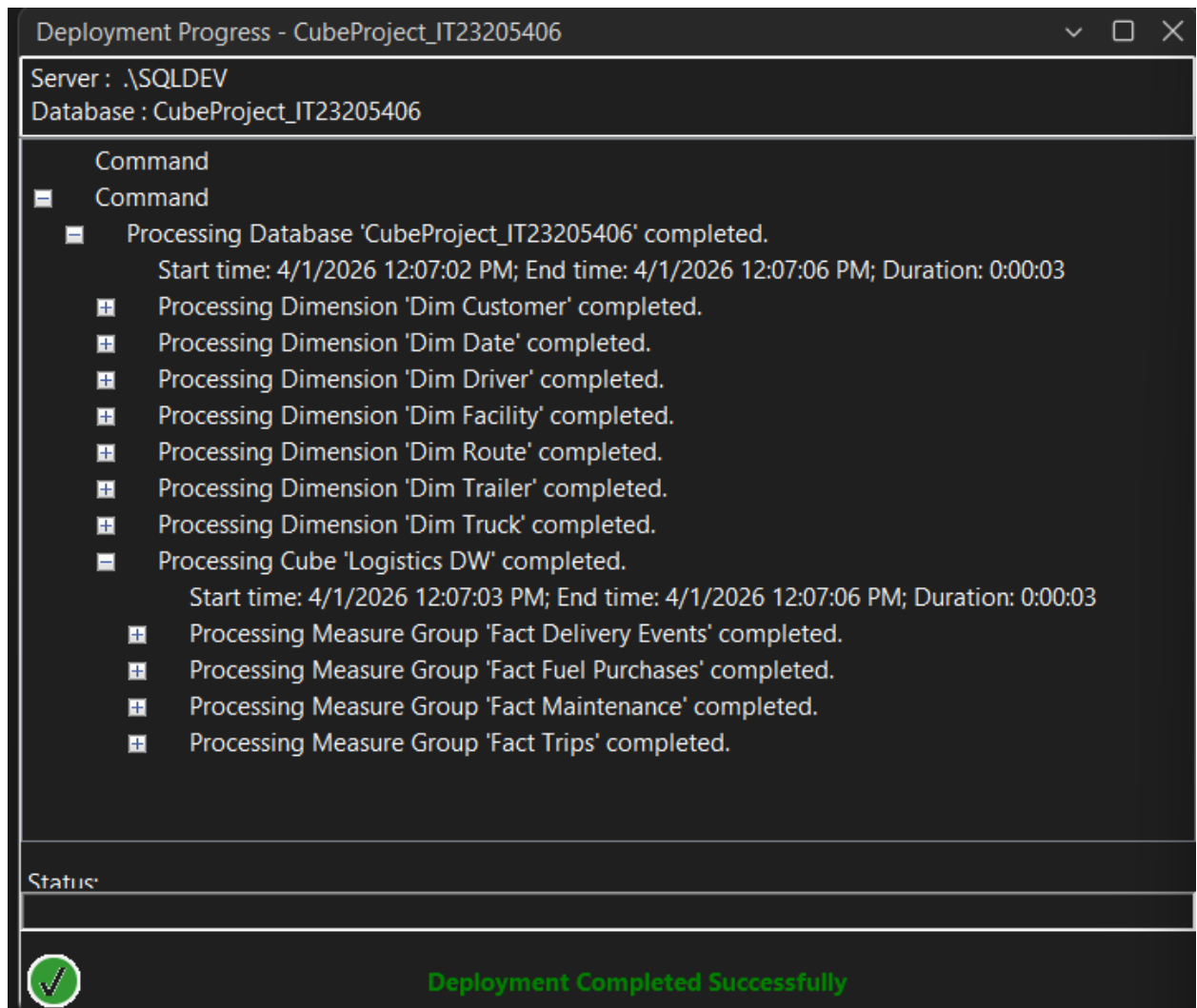


Figure 20 - Successfully deploying and processing the Logistics DW cube and all associated dimensions to the local SSAS server.

The processing phase executed successfully without encountering any attribute key-binding or relational mapping errors. This seamless execution ultimately confirmed the structural integrity and data cleanliness of the ETL pipeline established during the first phase of the assignment.

2.6 Key Performance Indicator (KPI) Development

To provide actionable business intelligence and evaluate the logistics network's performance against strategic goals, two custom Key Performance Indicators (KPIs) were developed within the SSAS multidimensional cube. These KPIs utilize Multidimensional Expressions (MDX) to calculate real-time performance metrics and translate them into intuitive visual scorecards for end-users.

To demonstrate a comprehensive understanding of MDX status logic, the KPIs were specifically designed to showcase two opposing evaluation metrics: a "Higher is Better" performance target and a "Lower is Better" financial constraint.

1. Delivery Volume Target (Performance Metric - Higher is Better)

- **Objective:** To monitor network throughput by tracking if the logistics operation is successfully hitting its target number of completed delivery events.
- **Value Expression:** [Measures].[Fact Delivery Events Count]
- **Goal Expression:** A target volume threshold was established to evaluate the actual measure against a baseline objective.
- **Status Indicator:** A standard "Traffic Light" visual was configured.
- **Status MDX Logic:** The status expression was written to return a value of 1 (Green) if the actual delivery volume met or exceeded the target goal. It returns 0 (Yellow) if the volume reached a minimum acceptable threshold of 85% of the goal, and -1 (Red) if the volume fell significantly below target.

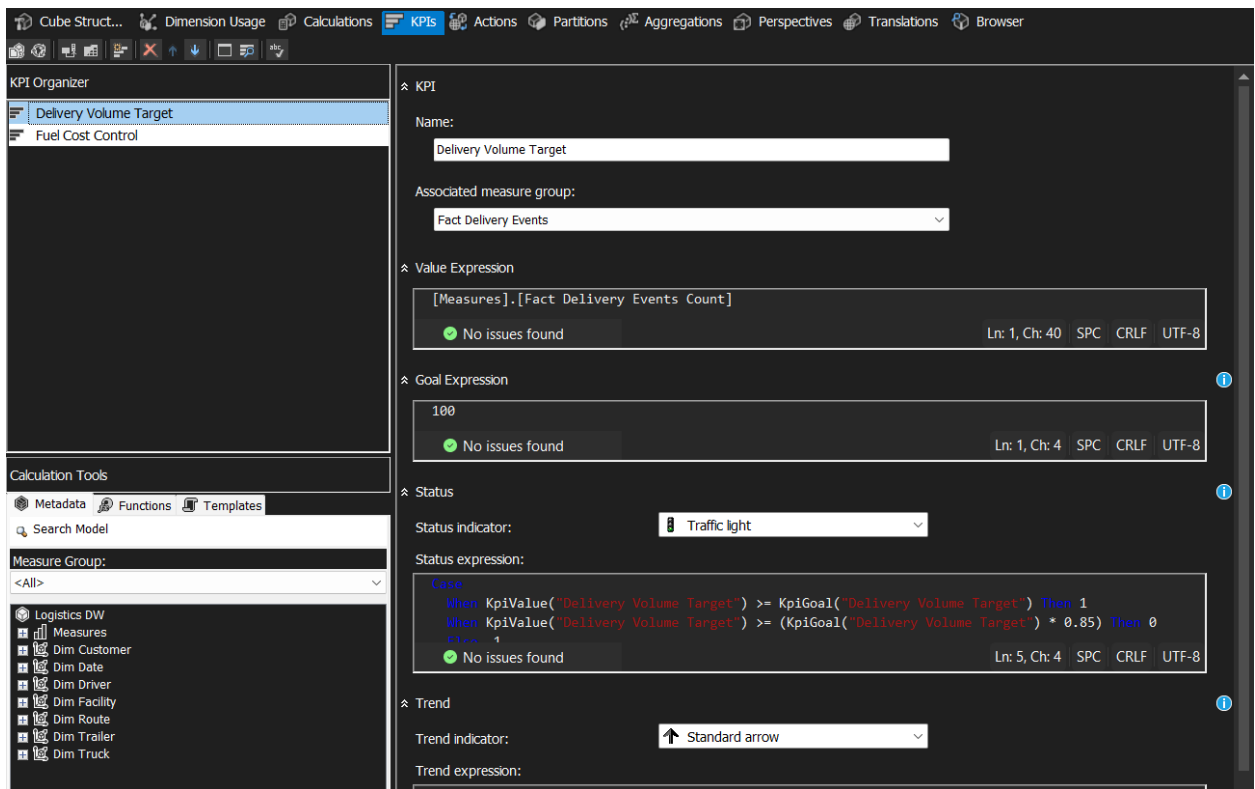


Figure 21 - Defining the 'Delivery Volume Target' Key Performance Indicator (KPI) with its associated Value, Goal, and Traffic Light Status expressions.

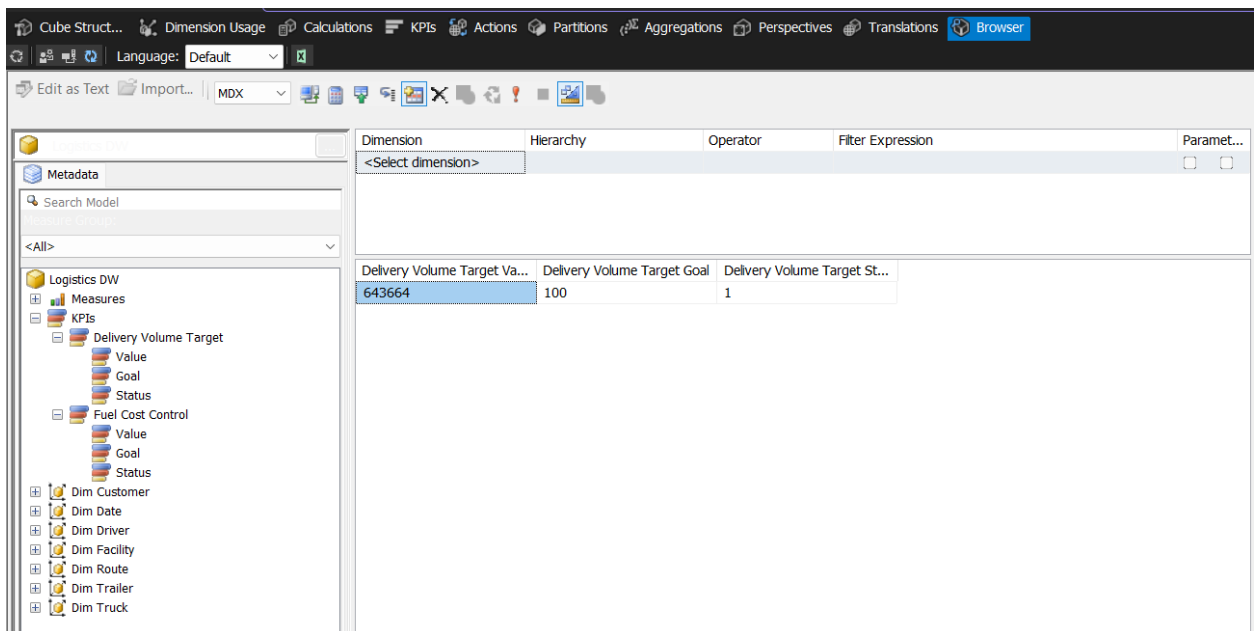


Figure 22 - Testing the 'Delivery Volume Target' KPI in the Browser view to ensure correct calculation of the target status.

2. Fuel Cost Control (Financial Metric - Lower is Better)

- **Objective:** To monitor operational expenses by tracking total fleet fuel expenditures against an allotted maximum budget.
- **Value Expression:** [Measures].[Cost] (derived from the Fact Fuel Purchases measure group).
- **Goal Expression:** A budgetary cap was established to serve as the benchmark.
- **Status Indicator:** A "Traffic Light" visual was utilized, requiring inverted logic to account for the financial constraint.
- **Status MDX Logic:** Because a lower cost represents a better outcome, the MDX CASE statement was inverted. The expression returns 1 (Green) if the actual cost remains strictly under or equal to the budget. It returns 0 (Yellow) if costs exceed the budget by no more than a 15% variance, and -1 (Red) if costs severely exceed the financial limit.

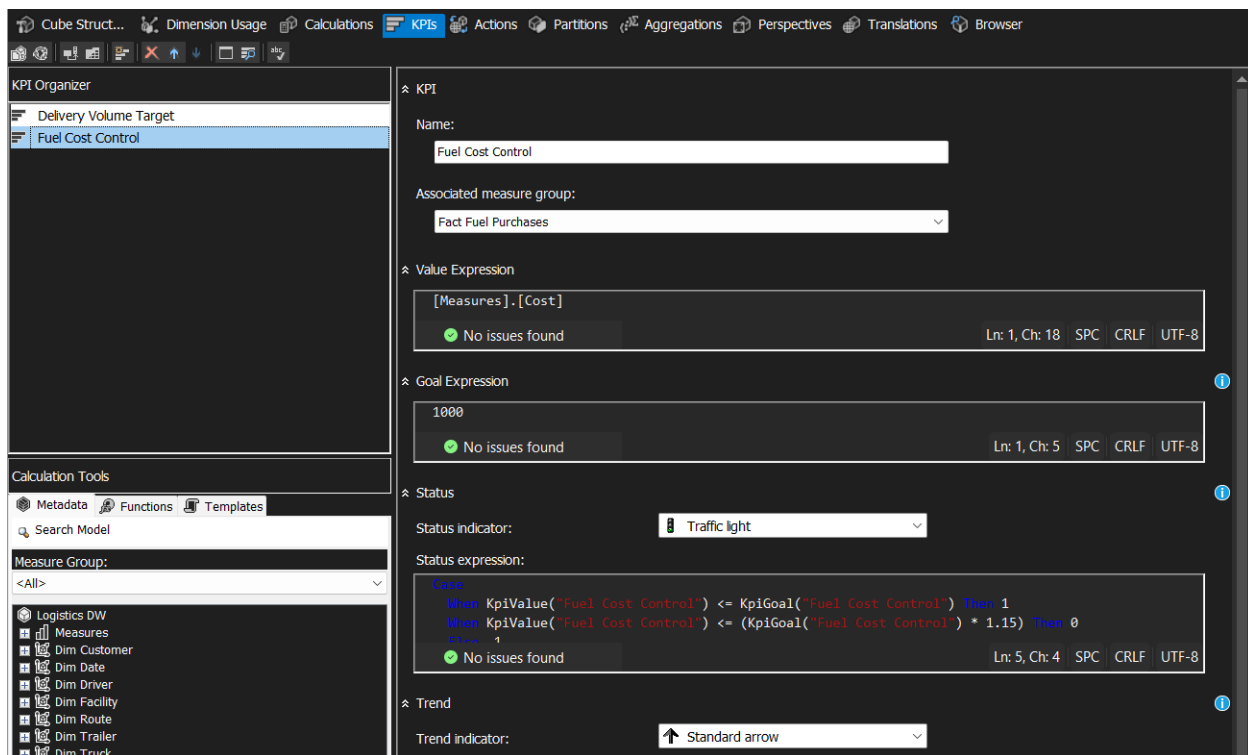


Figure 23 - Defining the 'Fuel Cost Control' Key Performance Indicator (KPI) to monitor fuel expenditure against a specified target.

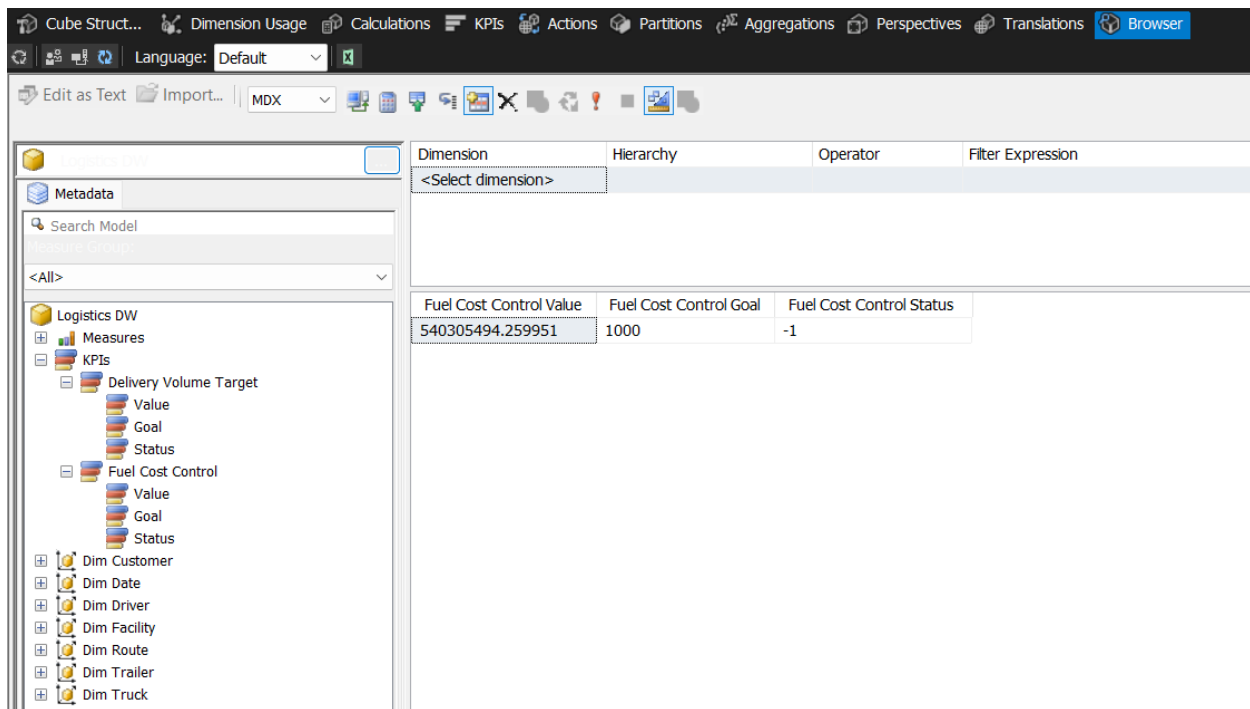


Figure 24 - Testing the 'Fuel Cost Control' KPI in the Browser view to verify the negative status trigger (-1).

Verification: Following the MDX scripting, the cube was re-deployed and processed. The KPIs were successfully verified using the integrated SSAS Browser View, confirming that the Traffic Light indicators accurately reflected the mathematical relationships between the Values and the established Goals.

Logistics DW.cube [Design] Dim Truck.dim [Design] Dim Facility.dim [Design] Dim Trailer.dim [Design] Dim Date.dim [Design]					
Cube Struct... Dimension Usage Calculations KPIs Actions Partitions Aggregations Perspectives Translations Browser					
Dimension Hierarchy Operator Filter Expression					
<Select dimension>					
Display Structure		Value	Goal	Status	Trend
Delivery Volume Target		643664	100		
Fuel Cost Control		540305494.26	1000		

Figure 25 - Final validation of both KPIs displaying their evaluated Traffic Light status indicators within the SSAS Browser interface.

Step 3: Demonstration of OLAP operations

3.1 Database Connection and Client Integration

To visualize the multidimensional data, Microsoft Excel was utilized as the primary front-end reporting tool. A live connection was established using the OLE DB provider for SQL Server Analysis Services. The workbook was connected directly to the local .\SQLDEV server instance, pointing specifically to the deployed Logistics DW cube. This live connection ensures that any future data processed in the Data Warehouse will immediately reflect in the Excel PivotTables without requiring manual data extracts.

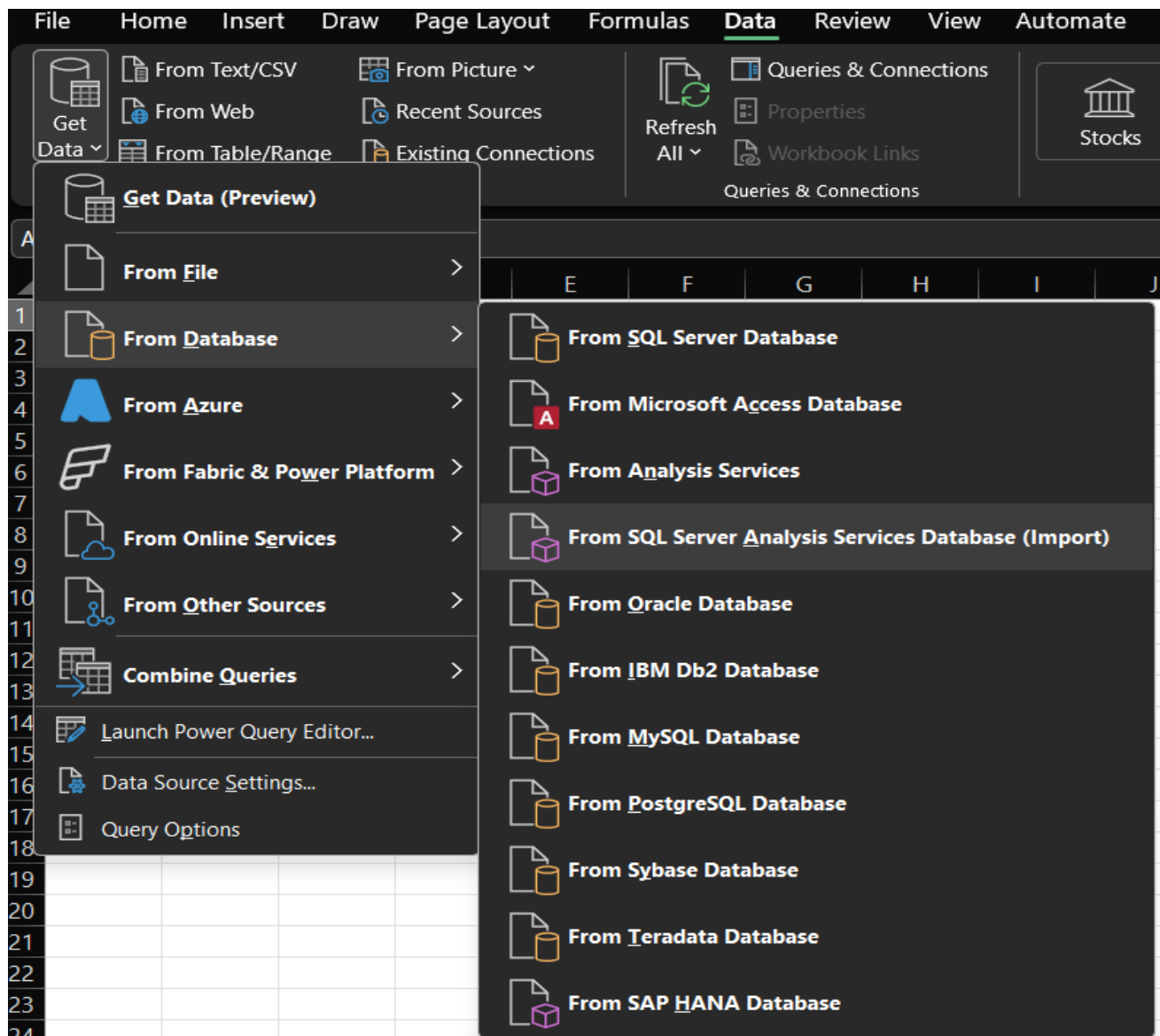
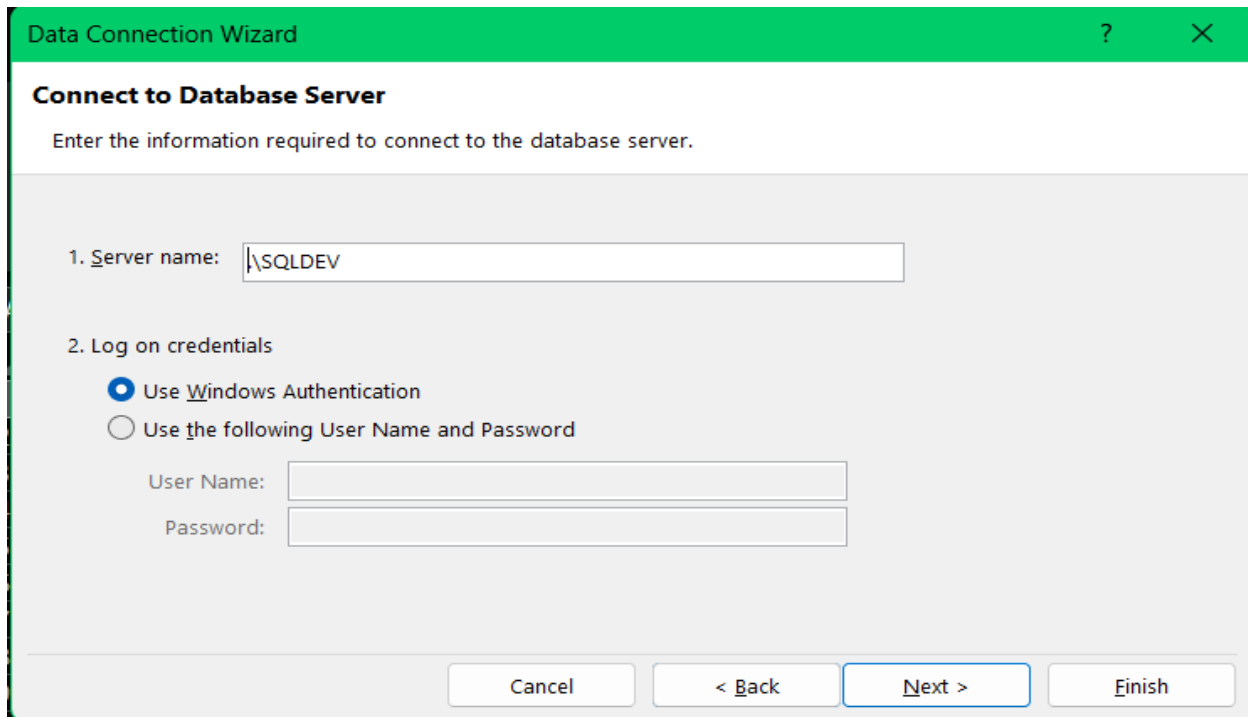


Figure 26 - Initiating the connection from Microsoft Excel to the deployed Analysis Services cube using the 'Get Data' feature.



Data Connection Wizard

Connect to Database Server

Enter the information required to connect to the database server.

1. Server name: \SQLDEV

2. Log on credentials

☒ Use Windows Authentication

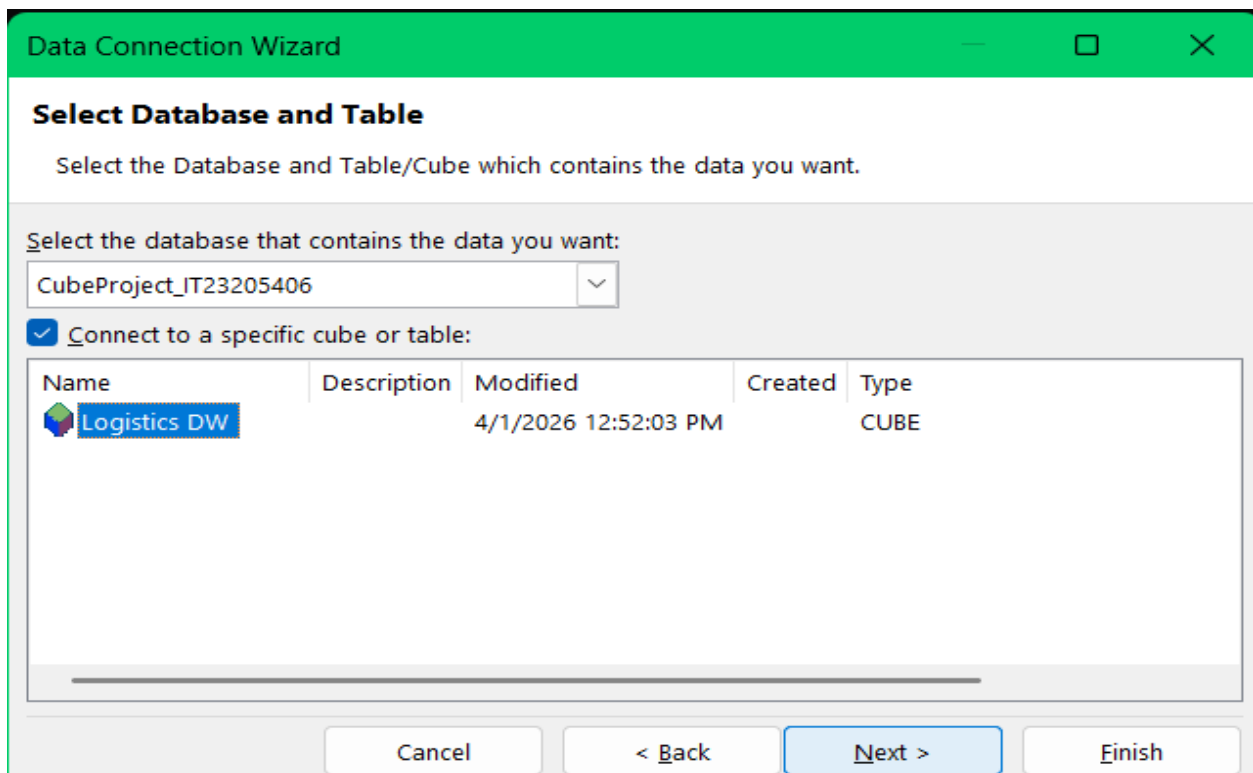
☐ Use the following User Name and Password

User Name:

Password:

Cancel < Back Next > Finish

Figure 27 - Configuring the Data Connection Wizard in Excel to authenticate and connect to the local .\SQLDEV database server.



Data Connection Wizard

Select Database and Table

Select the Database and Table/Cube which contains the data you want.

Select the database that contains the data you want:

CubeProject_IT23205406

☒ Connect to a specific cube or table:

Name	Description	Modified	Created	Type
 Logistics DW		4/1/2026 12:52:03 PM		CUBE

Cancel < Back Next > Finish

Figure 28 - Selecting the deployed Logistics DW cube from the Data Connection Wizard to import its multidimensional structure into Excel.

Save Data Connection File and Finish

Enter a name and description for your new Data Connection file, and press Finish to save.

File Name:
._SQLDEV CubeProject_IT23205406 Logistics DW.odc Browse...

☐ Save password in file

Description:
(To help others understand what your data connection points to)

Friendly Name:
._SQLDEV CubeProject_IT23205406 Logistics DW

Search Keywords:

☐ Always attempt to use this file to refresh data

Excel Services: Authentication Settings...

Cancel < Back Next > Finish

Figure 29 - Saving the Office Data Connection (.odc) file to finalize the secure Live Connection between Excel and the local SSAS instance.

3.2 Demonstration of OLAP Operations

Using the integrated Excel PivotTable feature, the following core Online Analytical Processing (OLAP) operations were executed to demonstrate the multidimensional capabilities of the underlying hierarchies:

1. Drill-Down

Drill-down involves navigating from a highly aggregated level of data to a more detailed, granular level.

- **Action Performed:** Utilizing the Geography hierarchy on the Rows axis, the top-level Country member was expanded to reveal the State/Province members underneath it. This allowed for a more detailed analysis of Delivery Volumes by specific regions rather than a national summary.

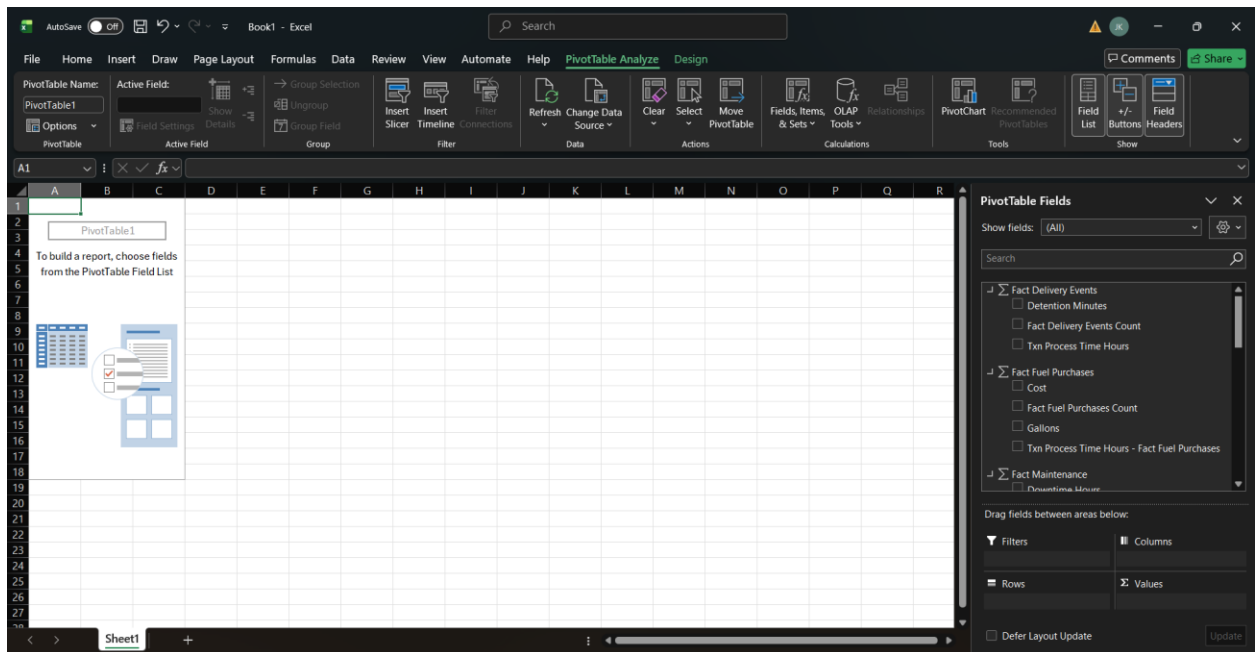


Figure 30 - The initialized Excel PivotTable successfully connected to the SSAS Cube, displaying the pre-calculated Measures and Dimensions in the Fields pane.

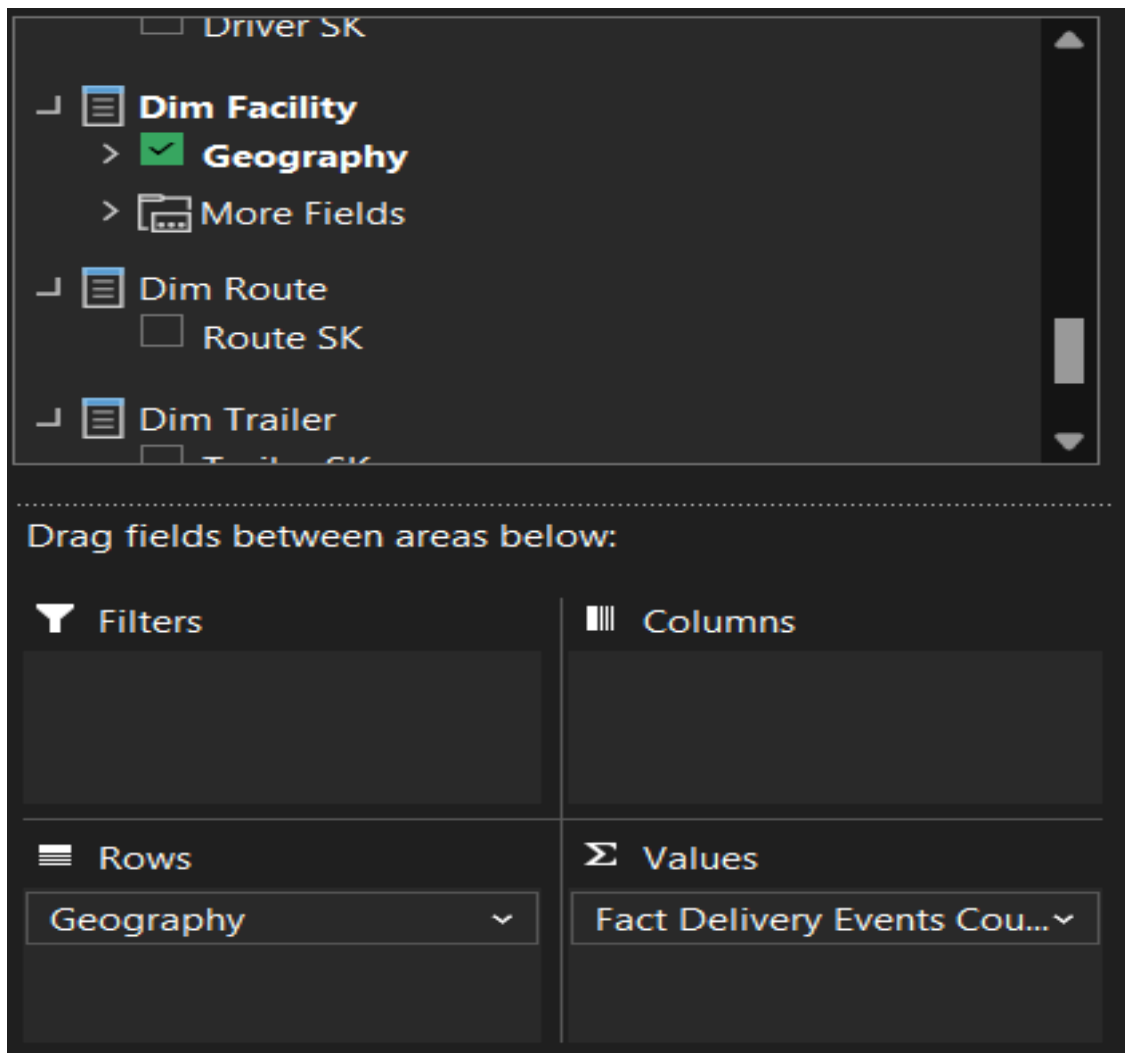


Figure 31 - Configuring the PivotTable by placing the Geography hierarchy into Rows and Fact Delivery Events Count into Values for spatial analysis.

A	B	C
Row Labels	Fact Delivery Events Count	
[-] USA		
+ AZ	6433	
[-] CA		
[-] Los Angeles		
FAC00031	3307	
FAC00046	3248	
+ CO	3243	
+ FL	9631	
+ GA	12747	
+ IL	3185	
+ IN	12948	
+ MI	12877	
+ MO	3228	
+ NC	9760	
+ NE	6447	
+ NV	9772	
+ NY	3229	
+ OK	3289	
+ OR	6301	
+ PA	6484	
+ TN	16067	
+ TX	15817	
+ UT	6382	
+ WI	6521	
Grand Total	160916	

Figure 32 - Demonstrating the **Drill-Down** OLAP operation by expanding the Geography hierarchy from the Country level down through States, Cities, and specific Facilities.

2. Roll-Up

Roll-up (or consolidation) is the exact inverse of a drill-down, where granular data is aggregated into a broader summary level.

- **Action Performed:** The StateProvince and City levels within the PivotTable were collapsed back up to the Country level. This effectively summed the Delivery Volume metrics into a single, high-level executive view.

Row Labels	Fact Delivery Events Count
USA	160916
Grand Total	160916

Figure 33 - A complete Roll-Up OLAP operation, collapsing the entire hierarchy to show the highest level of geographic aggregation (USA Grand Total).

3. Slice

A slice operation filters the entire multidimensional cube by a single member of a single dimension, effectively viewing one "page" of the cube.

- **Action Performed:** The Calendar Time dimension was added to the Filters pane. The data was filtered to isolate a single year. This sliced the cube to show geographic delivery performance exclusively for that specific time period.

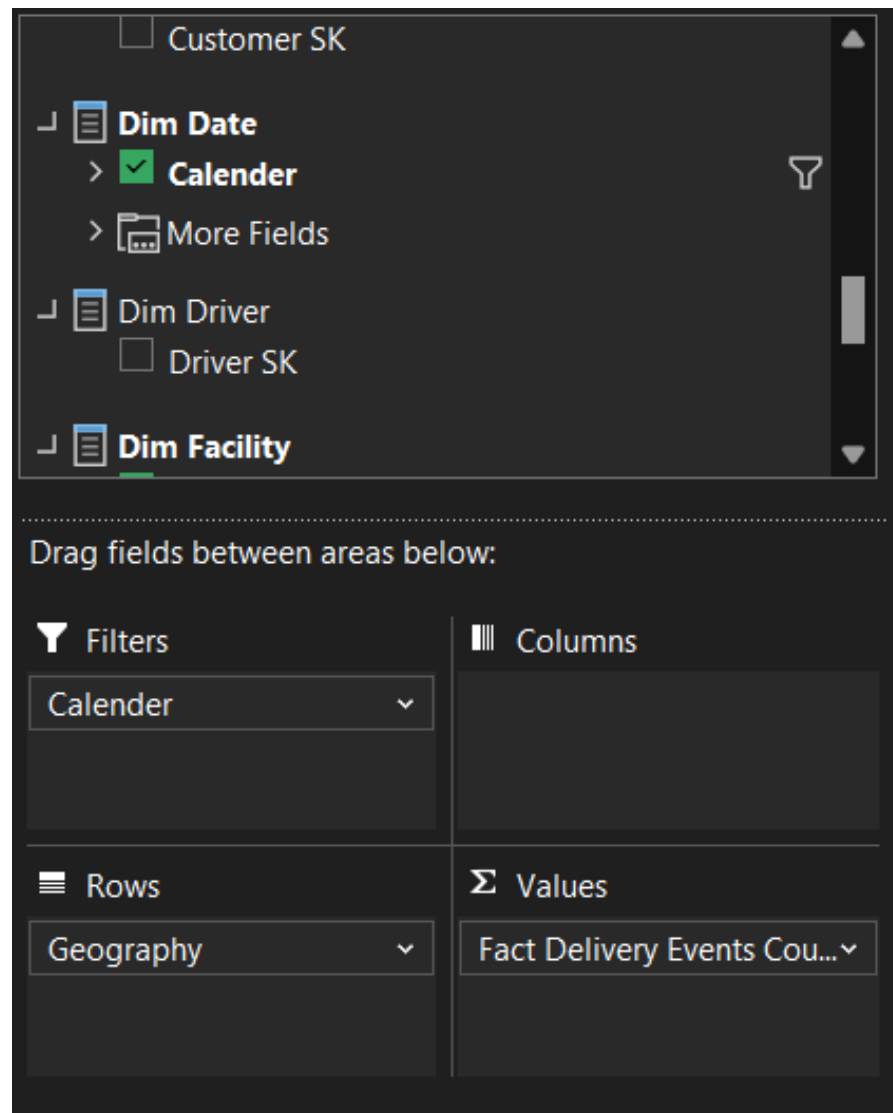


Figure 34 - Adding the Calender dimension to the Filters area to prepare for Slicing the dataset across different time periods.

Calender	2024	
Row Labels		Fact Delivery Events Count
USA		
AZ		2128
CA		2177
CO		1086
FL		3184
GA		4212
IL		1103
IN		4371
MI		4358
MO		1108
NC		3302
NE		2231
NV		3229
NY		1052
OK		1065
OR		2090
PA		2202
TN		5396
TX		5316
UT		2141
WI		2135
Grand Total		53886

Figure 35 - Demonstrating the **Slice** OLAP operation by filtering the multidimensional cube to isolate delivery event data exclusively for the year 2024.

Calender	2023
Row Labels	Fact Delivery Events Count
USA	
AZ	2133
CA	2188
CO	1092
FL	3183
GA	4284
IL	1042
IN	4261
MI	4164
MO	1029
NC	3204
NE	2112
NV	3251
NY	1076
OK	1119
OR	2097
PA	2064
TN	5330
TX	5251
UT	2098
WI	2170
Grand Total	53148

Figure 36 - Performing an additional **Slice** operation to dynamically pivot the view and analyze delivery event volumes specifically for the year 2023.

4. Dice

A dice operation isolates a specific "sub-cube" by applying filters across two or more dimensions simultaneously.

- **Action Performed:** Building upon the previous slice, the Fleet dimension was added to the Filters pane. The data was filtered to a specific Make of truck, while maintaining the specific Year filter. This provided a highly specific cross-section of data: analyzing deliveries in a specific region, during a specific year, utilizing a specific truck brand.

Truck	Peterbilt	
Calender	2024	
Row Labels	Fact Delivery Events Count	
USA		
AZ	2128	
CA	2177	
CO	1086	
FL	3184	
GA	4212	
IL	1103	
IN	4371	
MI	4358	
MO	1108	
NC	3302	
NE	2231	
NV	3229	
NY	1052	
OK	1065	
OR	2090	
PA	2202	
TN	5396	
TX	5316	
UT	2141	
WI	2135	
Grand Total	53886	

Figure 37 - Demonstrating the **Dice** OLAP operation by applying multiple filters simultaneously, creating a sub-cube of data isolated to 'Peterbilt' trucks during the year '2024'.

Truck	Peterbilt	
Calender	2025	
Row Labels		Fact Delivery Events Count
- USA		
+ AZ		5
+ CA		1
+ CO		1
+ FL		7
+ GA		5
+ IL		2
+ IN		4
+ MI		10
+ MO		2
+ NC		8
+ NE		4
+ NV		3
+ OR		7
+ PA		4
+ TN		12
+ TX		7
+ UT		2
+ WI		2
Grand Total		86

Figure 38 - Executing a secondary **Dice** operation by updating the time dimension parameter to '2025' while retaining the 'Peterbilt' vehicle filter, instantly recalculating the geographic sub-cube.

5. Pivot

Pivoting (or rotating) involves swapping the axes of the data presentation to provide an alternative perspective on the identical dataset.

- **Action Performed:** The PivotTable axes were rotated by moving the Geography hierarchy from the Rows area to the Columns area, and moving the Calendar Time hierarchy into the Rows area. This instantly transformed the report layout, allowing for a side-by-side geographic comparison over time without altering the underlying filter context.

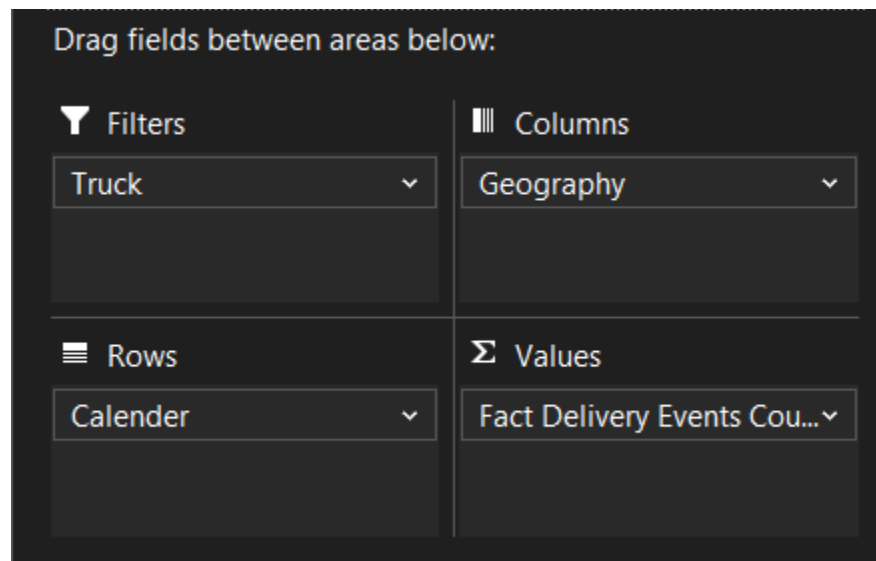


Figure 39 - Reconfiguring the PivotTable field mappings to prepare for a Pivot operation, specifically by moving the Geography hierarchy to Columns and the Calender hierarchy to Rows.

Fact Delivery Events Count		Column Labels																		Grand Total
		USA																		
		AZ																		
		Phoenix																		
		CA																		
		Los Angeles																		
		CO																		
		FL																		
		GA																		
		IL																		
		IN																		
		MI																		
		MO																		
		NC																		
		NE																		
		NV																		
		OR																		
		PA																		
		TN																		
		TX																		
		UT																		
		WI																		
Row Labels	FAC00015	FAC00035	FAC00046																	
# 2025	3	2	1	1	7	5	2	4	10	2	8	4	3	7	4	12	7	2	2	86
Grand Total	3	2	1	1	7	5	2	4	10	2	8	4	3	7	4	12	7	2	2	86

Figure 40 - Demonstrating the **Pivot** OLAP operation. Rotating the data axes transforms the vertical layout into a horizontal, cross-tabular matrix, providing a new perspective on delivery events across facilities.

Step 4: PowerBI Reports

4.1: Data Preparation & Modeling

For this project, data preparation and modeling were handled upstream using SQL Server Integration Services (SSIS) to populate a Galaxy Schema in SQL Server, followed by the deployment of a SQL Server Analysis Services (SSAS) Cube (Logistics_DW).

To build the Power BI reports, a **Live Connection** was established to the local SSAS instance. Because of this Live Connection:

- **Modeling:** No additional relationships or data modeling were required in Power BI. The tool inherently recognized the facts, dimensions, and hierarchical structures defined in the Cube.
- **DAX:** Explicit DAX queries were not necessary for these base reports, as the underlying measures (e.g., Fact Delivery Events Count) and aggregations were pre-calculated and imported directly from the Analysis Services Measure Groups.

4.2 : Initial Setup: Connecting to the SSAS Cube

Before building the reports, I established a Live Connection between Power BI Desktop and the local SQL Server Analysis Services (SSAS) instance. This method ensures that Power BI inherits all the pre-built hierarchies, relationships, and measures directly from the cube without needing to import the raw data.

Connection Steps:

1. **Select Data Source:** In Power BI Desktop, I clicked **Get Data**, searched for "Analysis Services," and selected the **SQL Server Analysis Services database** connector.

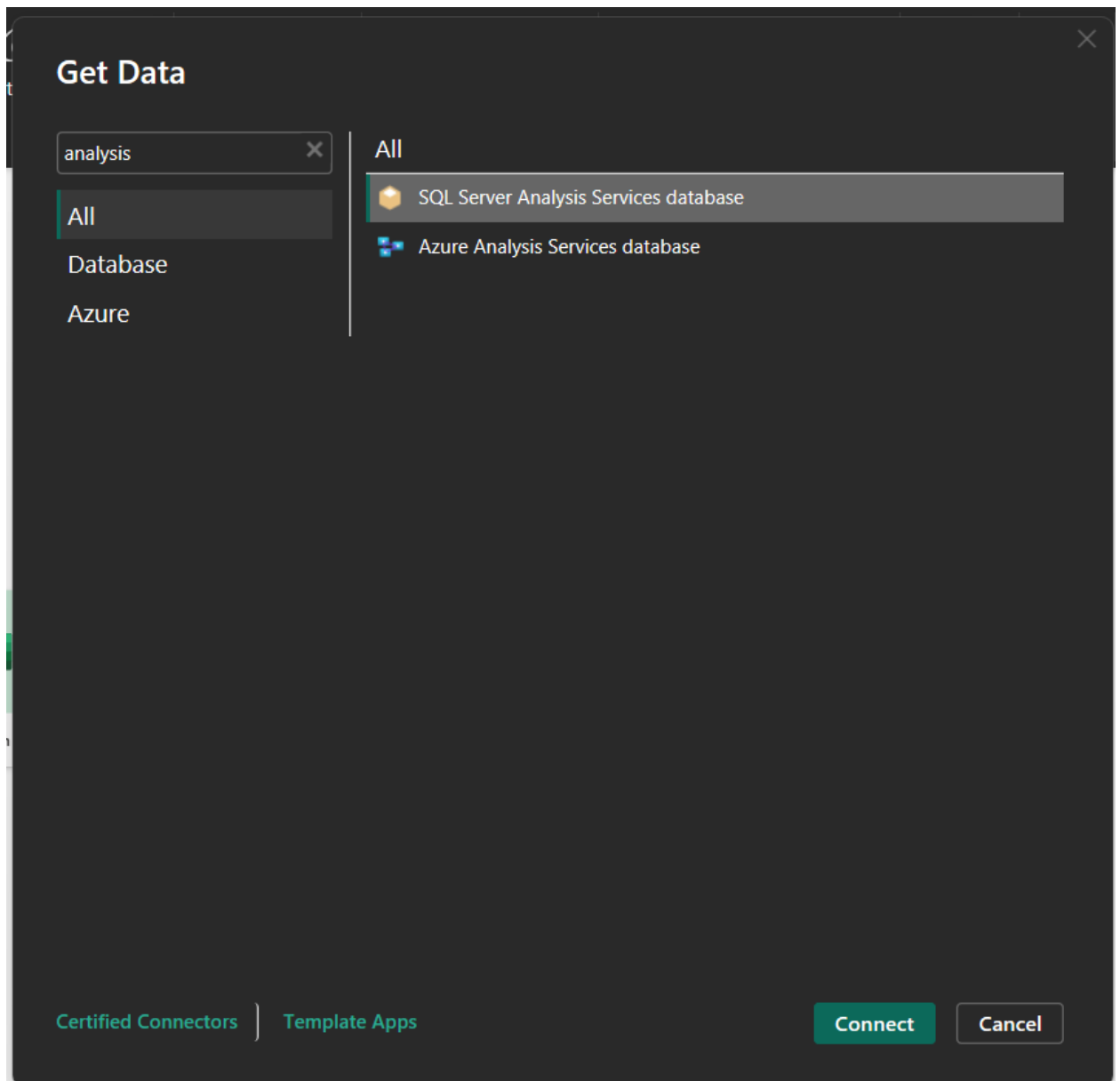


Figure 41 - Selecting the SQL Server Analysis Services database connector.

2. **Server Configuration:** In the database dialog prompt, I entered `.\SQLDEV` as the local server name. I explicitly selected the **Connect live** radio button under the data connectivity mode options to maintain a direct query path to the cube.

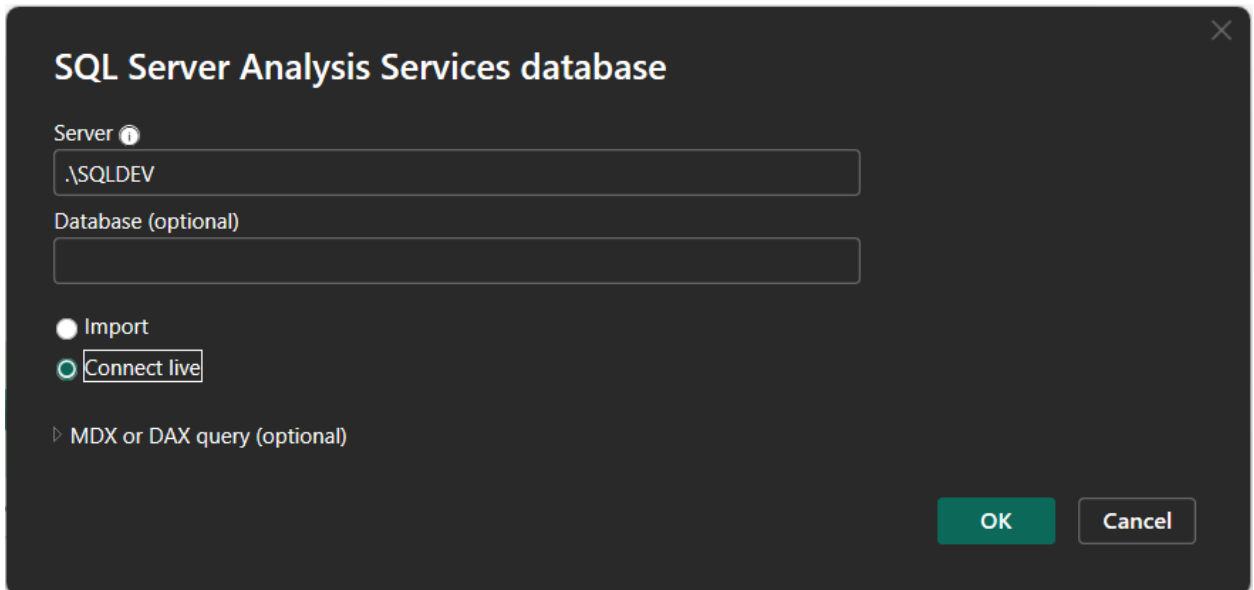


Figure 42 - Configuring the Live Connection to the local `.\SQLDEV` server.

3. **Authentication:** I used Windows Authentication, selecting **"Use my current credentials"** to grant Power BI access to the local SSAS instance.

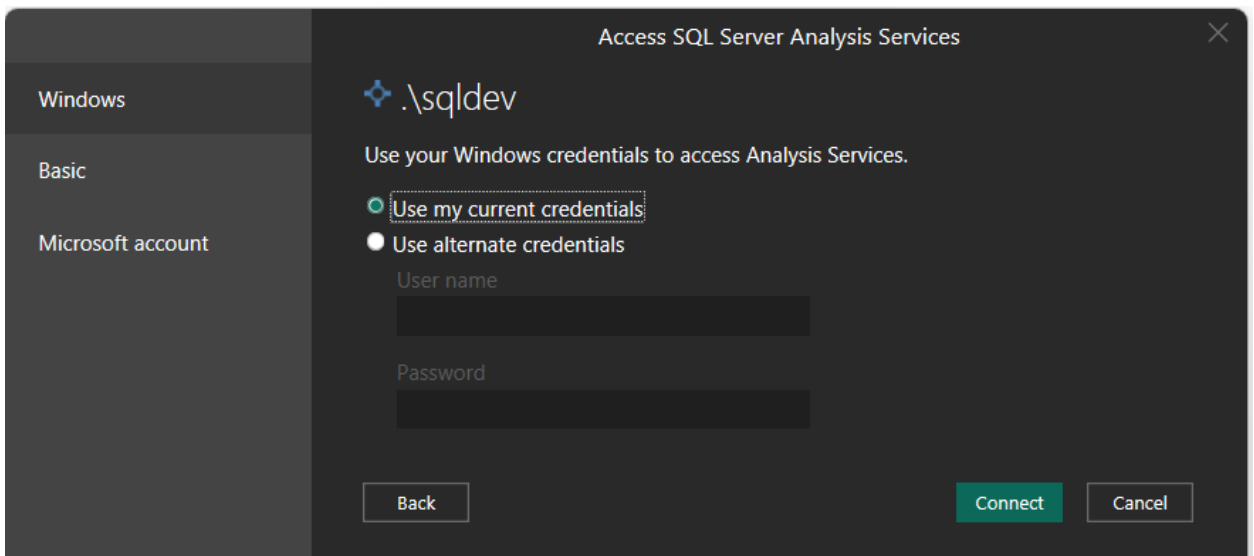


Figure 43 - Authenticating the connection using Windows credentials.

4. **Selecting the Cube:** In the Navigator window, I expanded the local server and the deployed project folder (CubeProject_IT23205406). I selected the **Logistics DW** cube, which displayed a preview of the available dimensions (e.g., Dim Customer, Dim Date) and measures (e.g., Fact Delivery Events Count).

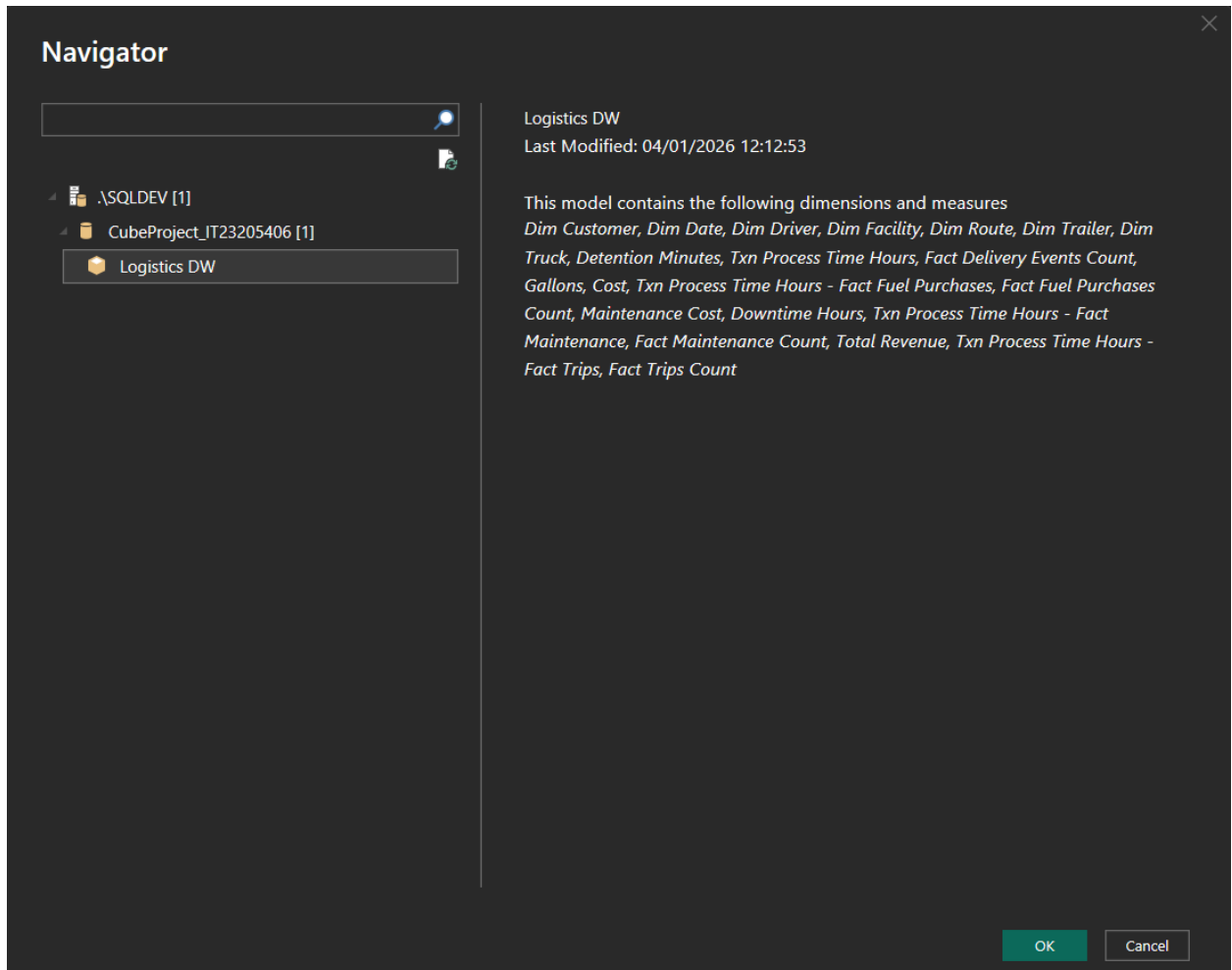


Figure 44 - Selecting the Logistics DW cube from the Navigator window.

5. **Load:** After clicking **OK**, the pre-modeled Fact and Dimension tables successfully populated in the Data pane, ready for visualization.

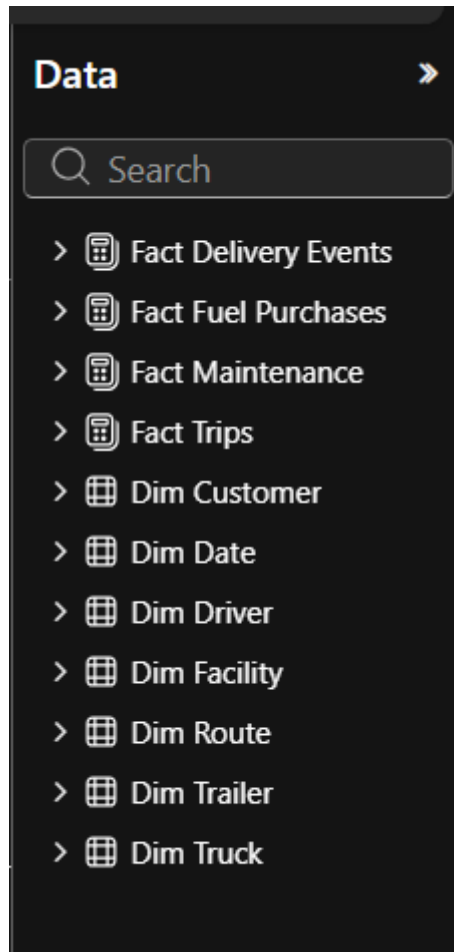


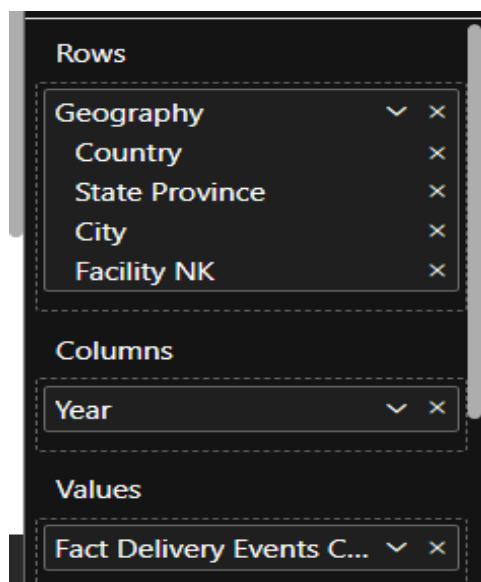
Figure 45 - The Data pane successfully populated with SSAS Dimension and Fact tables.

4. 3: Report 1: Matrix Visual with Groupings

- **Objective:** Display detailed tabular data with row and column groupings.
- **Visual Design:** A Matrix visual was selected to provide a clean, Excel-like pivot layout.
- **Steps Followed:**
 1. Inserted a **Matrix** visual onto a blank canvas.



2. Placed the pre-defined **Geography Hierarchy** (StateProvince > City) into the **Rows** field well to allow for row-level expansion.
3. Placed the **Year** attribute from the DimDate table into the **Columns** field well.
4. Placed the **Fact Delivery Events Count** measure into the **Values** field well.



- **Outcome:** The report successfully displays delivery volumes, allowing users to expand states into cities while viewing the data categorized by year across the columns.

Country	2022	2023	2024	2025	Total
USA	53796	53148	53886	86	160916
AZ	2167	2133	2128	5	6433
Phoenix	2167	2133	2128	5	6433
FAC00015	1079	1078	1032	3	3192
FAC00035	1088	1055	1096	2	3241
CA	2189	2188	2177	1	6555
Los Angeles	2189	2188	2177	1	6555
FAC00031	1100	1138	1069		3307
FAC00046	1089	1050	1108	1	3248
CO	1064	1092	1086	1	3243
FL	3257	3183	3184	7	9631
Miami	3257	3183	3184	7	9631
FAC00029	1074	1083	1045	3	3205
FAC00032	1080	1058	1030	2	3170
FAC00033	1103	1042	1109	2	3256
GA	4246	4284	4212	5	12747
IL	1038	1042	1103	2	3185
IN	4312	4261	4371	4	12948
MI	4345	4164	4358	10	12877
MO	1089	1029	1108	2	3228
NC	3246	3204	3302	8	9760
NE	2100	2112	2231	4	6447
NV	3289	3251	3229	3	9772
NY	1101	1076	1052		3229
OK	1105	1119	1065		3289
OR	2107	2097	2090	7	6301
PA	2214	2064	2202	4	6484
TN	5329	5330	5396	12	16067
TX	5243	5251	5316	7	15817
UT	2141	2098	2141	2	6382
WI	2214	2170	2135	2	6521
Total	53796	53148	53886	86	160916

Figure 46 - A comprehensive cross-tabular view resulting from a complex Pivot operation. This matrix successfully displays the fully expanded Geography hierarchy (down to the specific Facility level) analyzed side-by-side across multiple years (2022–2025), demonstrating the full multidimensional analytical power of the cube.

4. 4: Report 2: Multiple Visuals and Cascading Slicers

- **Objective:** Create an interactive dashboard with cascading filters and multiple charts to present insights from the data.
- **Visual Design:** Utilized a diverse set of visuals including a Clustered Bar Chart, a Donut Chart, a Treemap, and a Line Chart to analyze delivery event volumes across different geographic levels.
- **Steps Followed:**
 1. **Clustered Bar Chart:** Added this visual to show a straightforward ranking. Placed State Province (from Dim Facility) on the Y-axis and Fact Delivery Events Count on the X-axis.

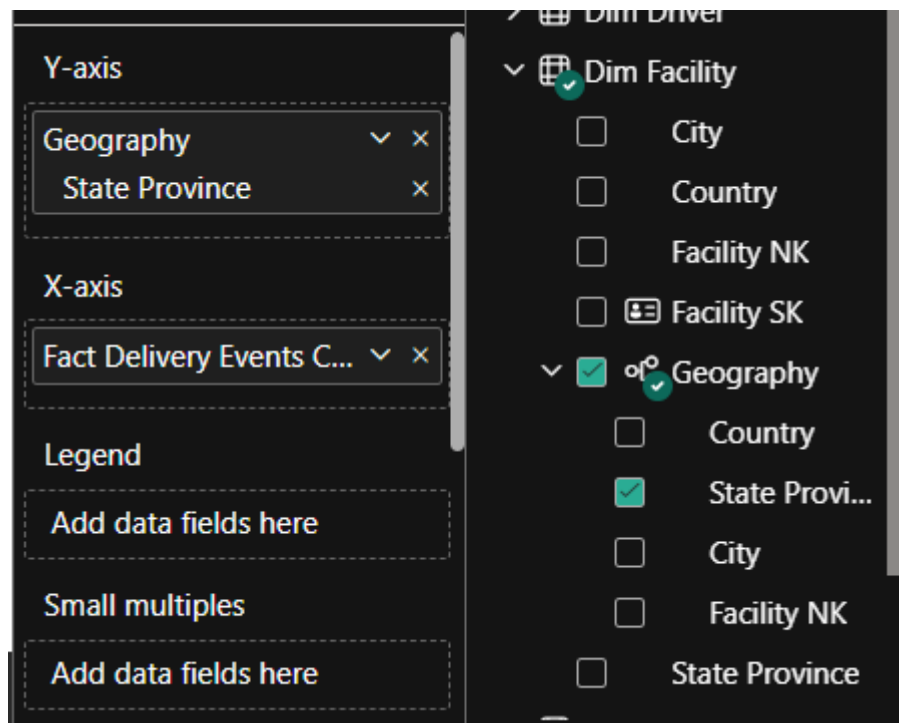


Figure 47 - Configuring the Clustered Bar Chart (Delivery Count by State Province).

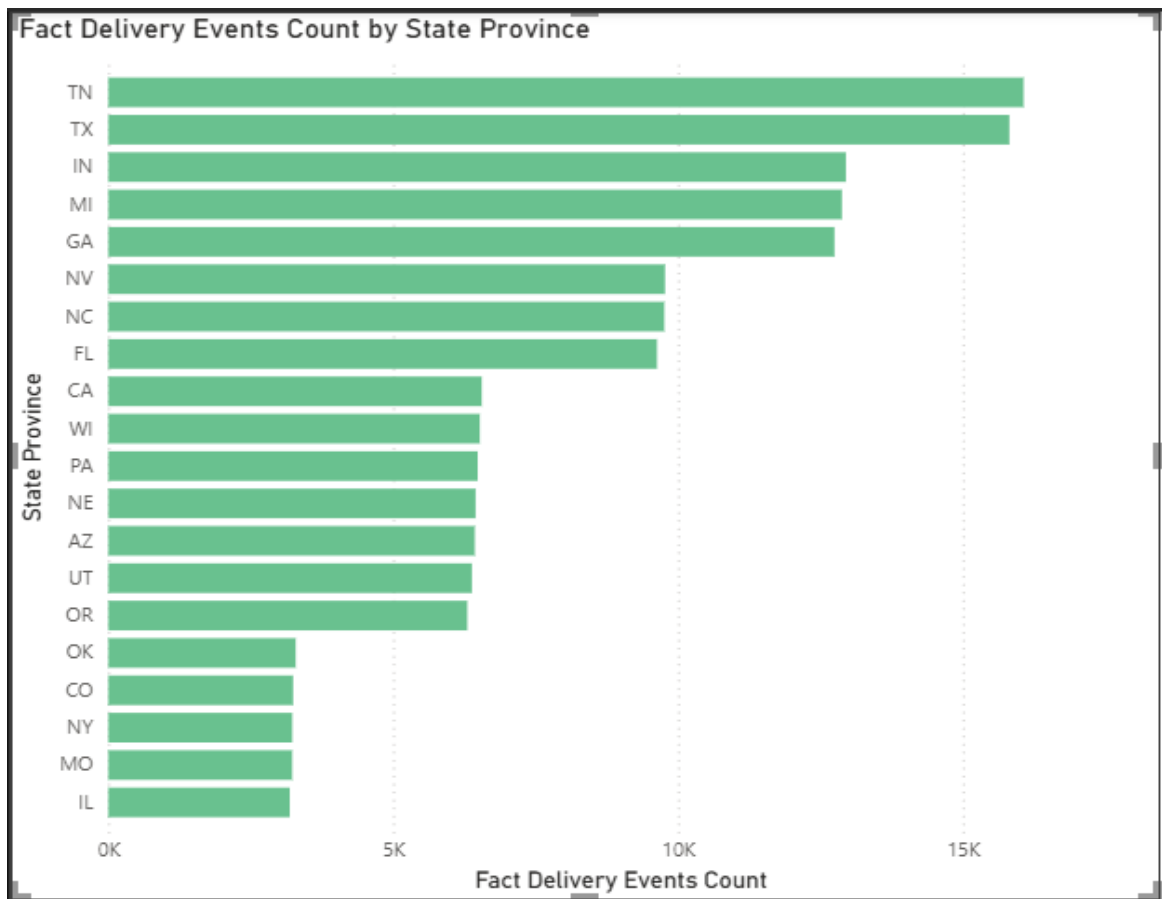


Figure 48 – The foundational Clustered Bar Chart created to rank delivery event volumes by State Province.

2. Donut Chart: Added this visual to show proportional distributions. Placed the Geography hierarchy (State Province and City) into the Legend well, and Fact Delivery Events Count into the Values well.

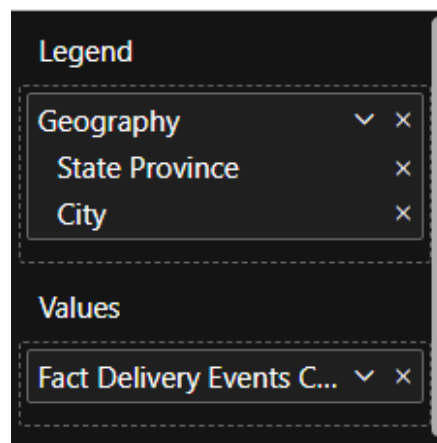


Figure 49 - Configuring the Donut Chart (Proportional distribution by Geography).

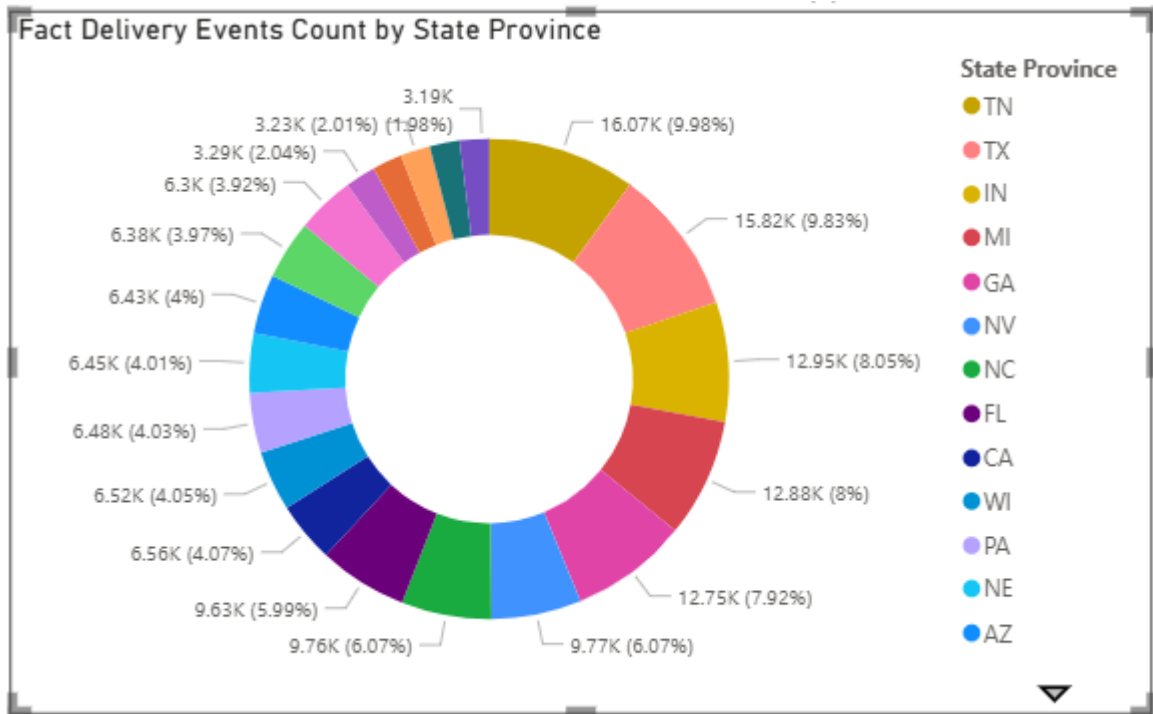


Figure 50 - A Donut Chart visualization displaying the proportional distribution of delivery events across the states.

3. Treemap: Added this visual to display relative volumes as spatial blocks. Placed the Geography hierarchy into the Category well, and Fact Delivery Events Count into the Values well.

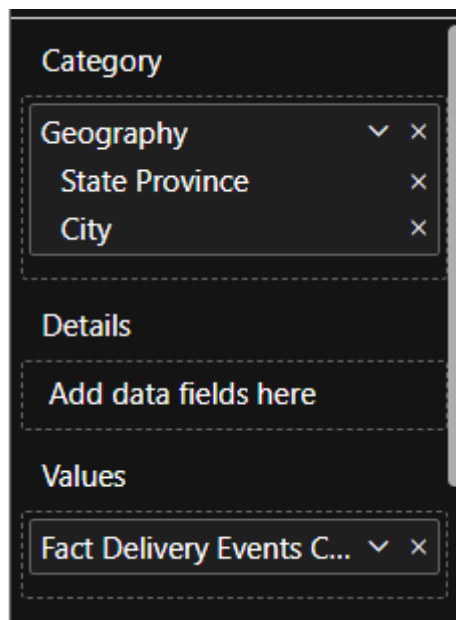


Figure 51 - Configuring the Treemap visual for relative volume comparison.

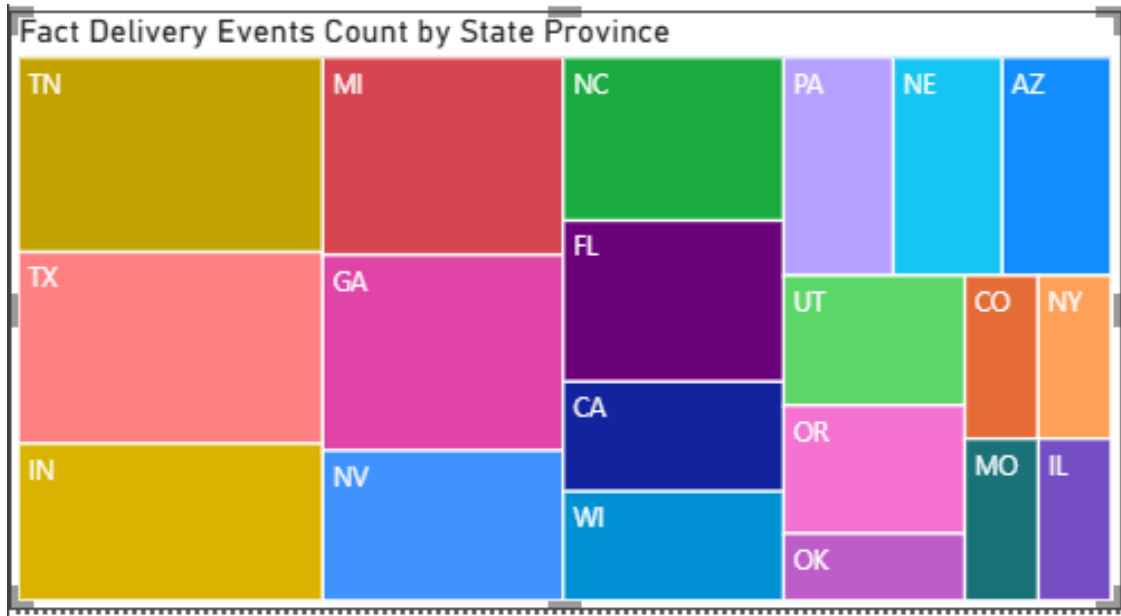


Figure 52 - A Treemap visual providing a block-based spatial comparison of state-level delivery volumes.

4. Line Chart: Added this visual to track variations across regions. Placed the Geography hierarchy on the X-axis and Fact Delivery Events Count on the Y-axis.

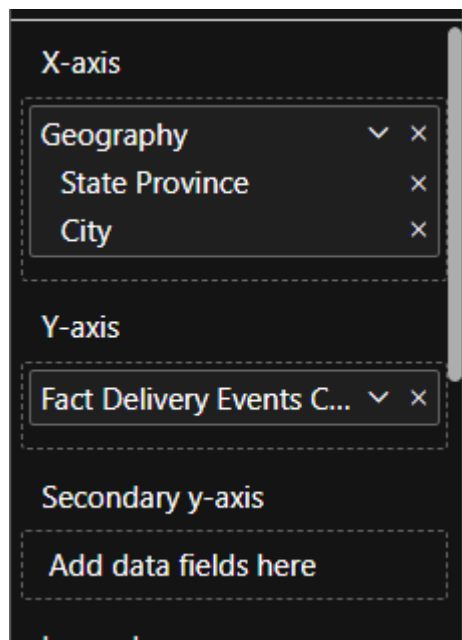


Figure 53 - Configuring the Line Chart (Tracking volume variations across cities).

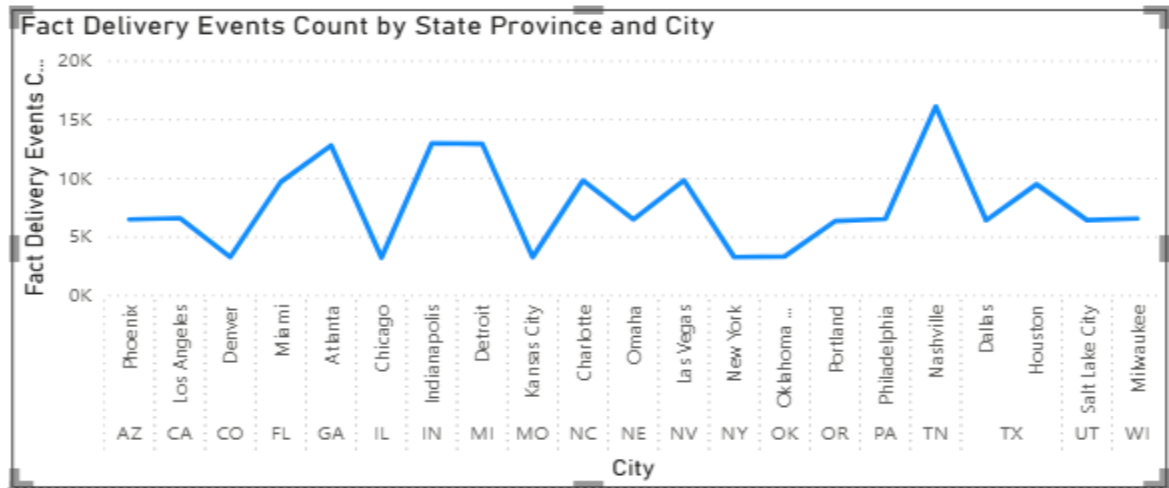


Figure 55 - A Line Chart tracking the variations in delivery event counts across individual cities.

5. Slicers (Parameters): Inserted a Slicer visual for StateProvince and a second Slicer visual for City to fulfill the parameter requirement.

Cascading Implementation: Because the SSAS data model defines the hierarchical relationship between State and City, selecting a state in the first slicer dynamically filters the second slicer to only show cities within that specific state. Both slicers successfully cross-filter all four visualizations (Bar, Donut, Treemap, and Line Chart) on the canvas.

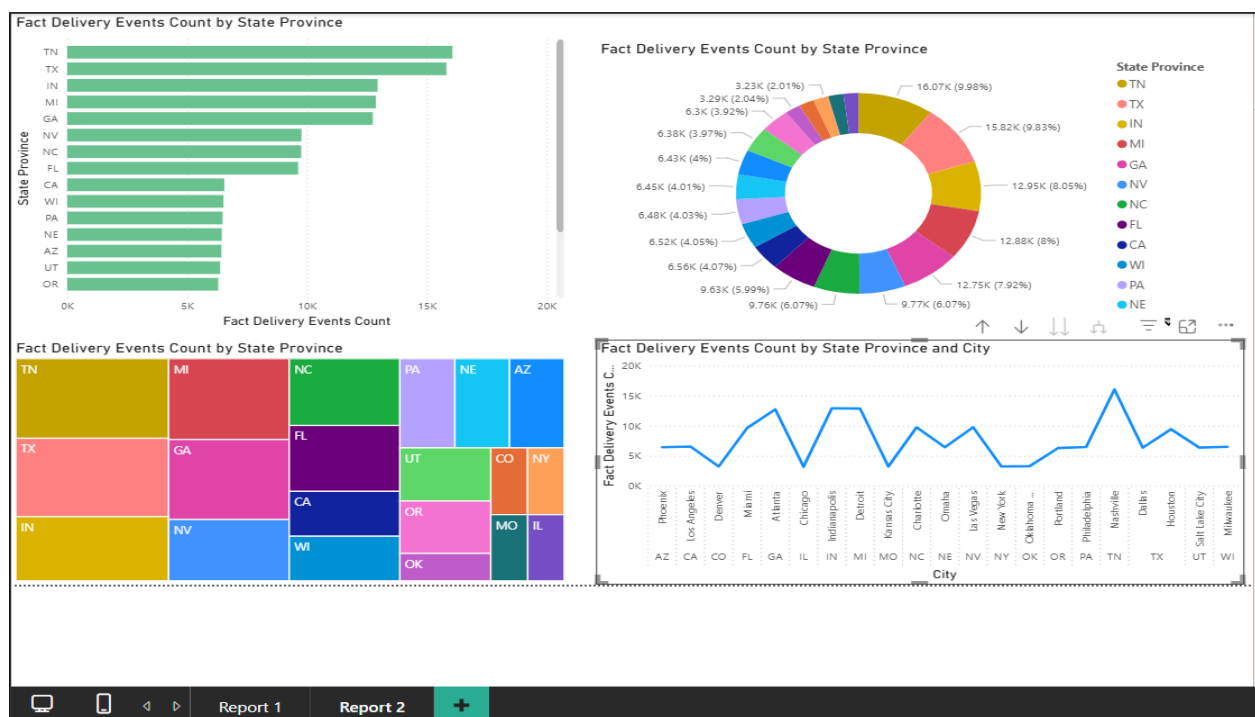


Figure 56 - The completed interactive dashboard for Report 2, combining all four geographical visuals into a single cohesive view.

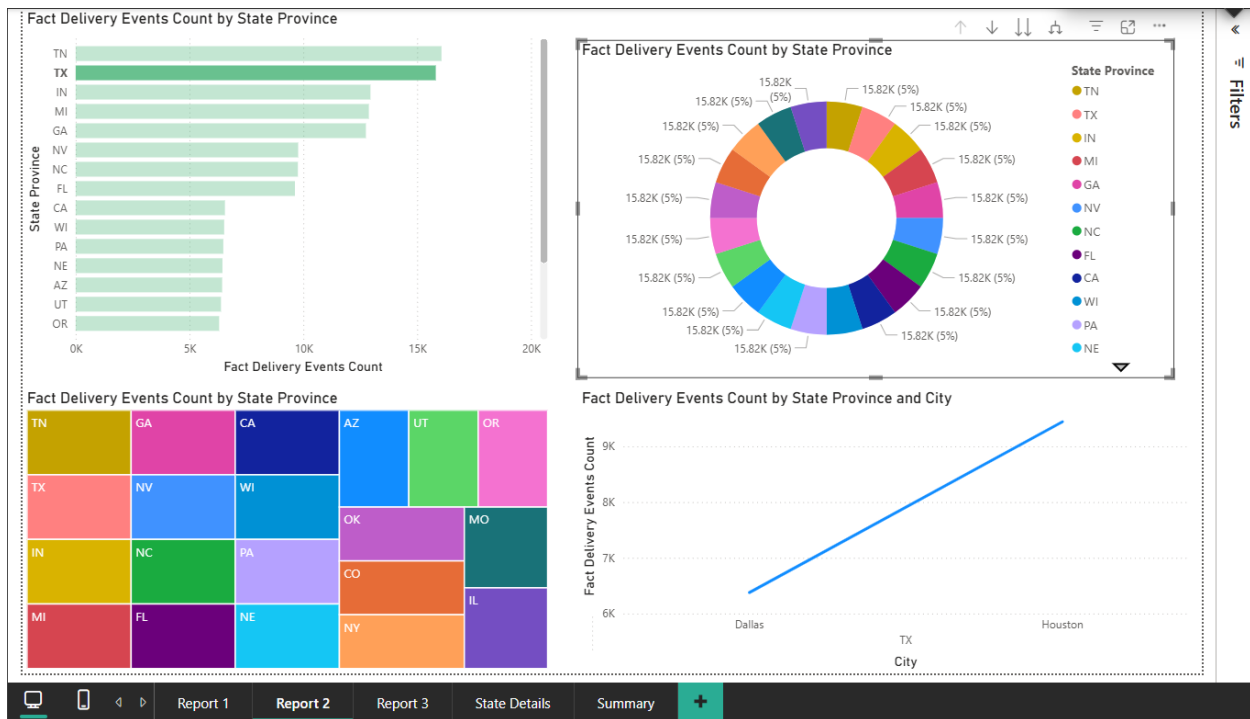


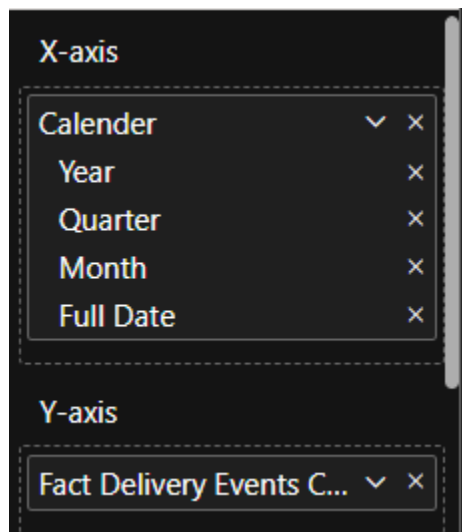
Figure 57 - Demonstrating native cross-filtering: Selecting 'TX' (Texas) in the bar chart dynamically filters the donut chart, treemap, and limits the line chart to only display Texas cities (Dallas and Houston).

4. 5: Report 3: Hierarchical Drill-Down Report

- **Objective:** Allow users to explore data hierarchically across time dimensions.
- **Visual Design:** A Clustered Column Chart was chosen, as vertical columns are the standard best practice for displaying time-series data.
- **Steps Followed:**
 1. Inserted a Clustered Column Chart onto the canvas.



2. Placed the complete Calender Hierarchy (Year > Quarter > Month > Full Date) from the Dim Date table onto the X-axis.
3. Placed the Fact Delivery Events Count measure onto the Y-axis.



4. Utilized the "Drill Down" and "Expand all down one level in the hierarchy" features (the arrow icons on the visual header) to navigate through the time levels.

- **Outcome:** Users can click on a specific aggregate time frame (like a Year or Quarter) to drill down into its respective sub-categories, all the way down to the granular Full Date level (as demonstrated by the expanded X-axis showing 2022-01-01, etc.). This successfully leverages the SSAS cube's native time hierarchy to provide deep temporal analysis without writing custom DAX.

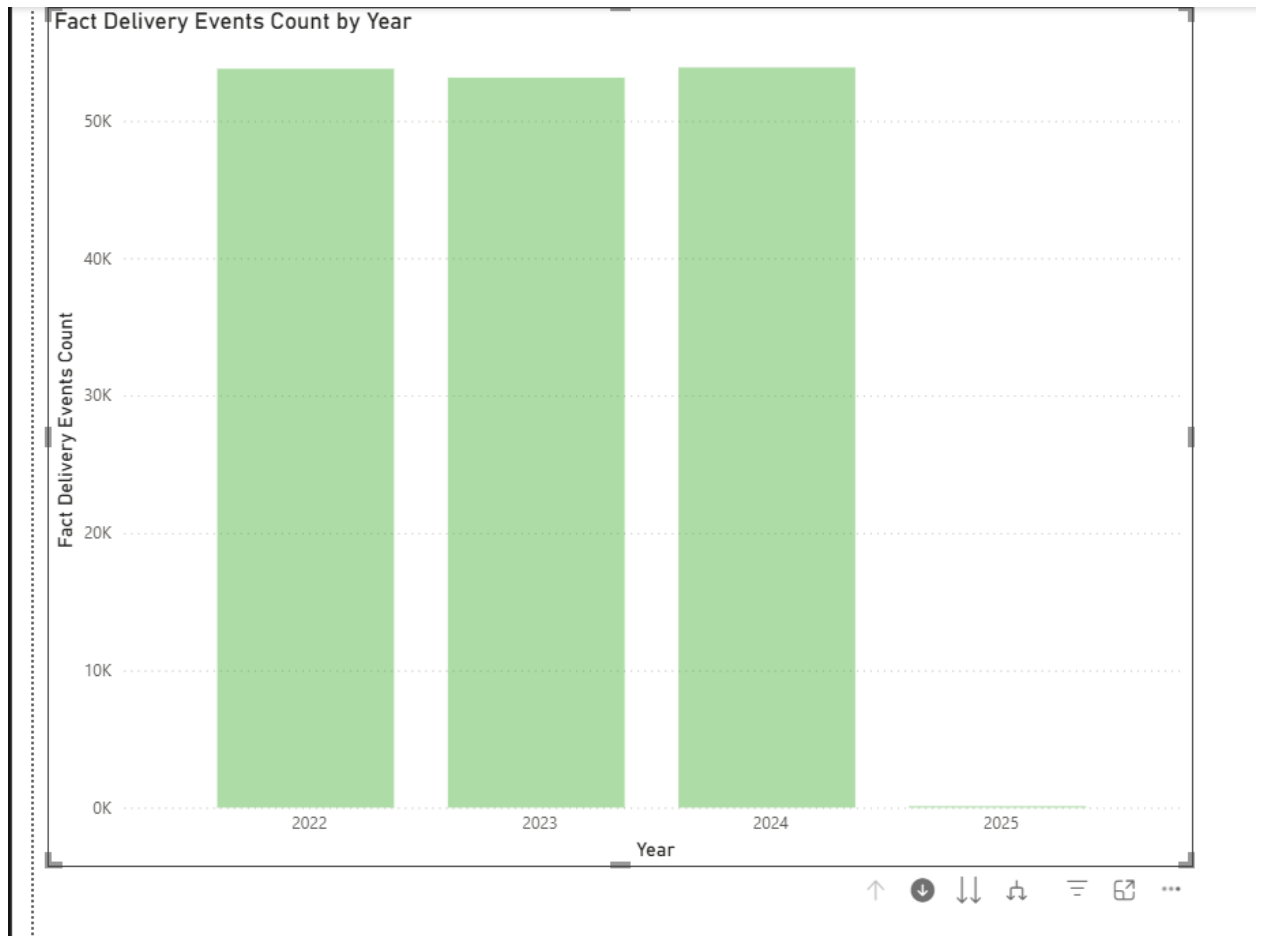


Figure 58 - The top level of the temporal drill-down, displaying aggregated delivery event counts summarized by Year.

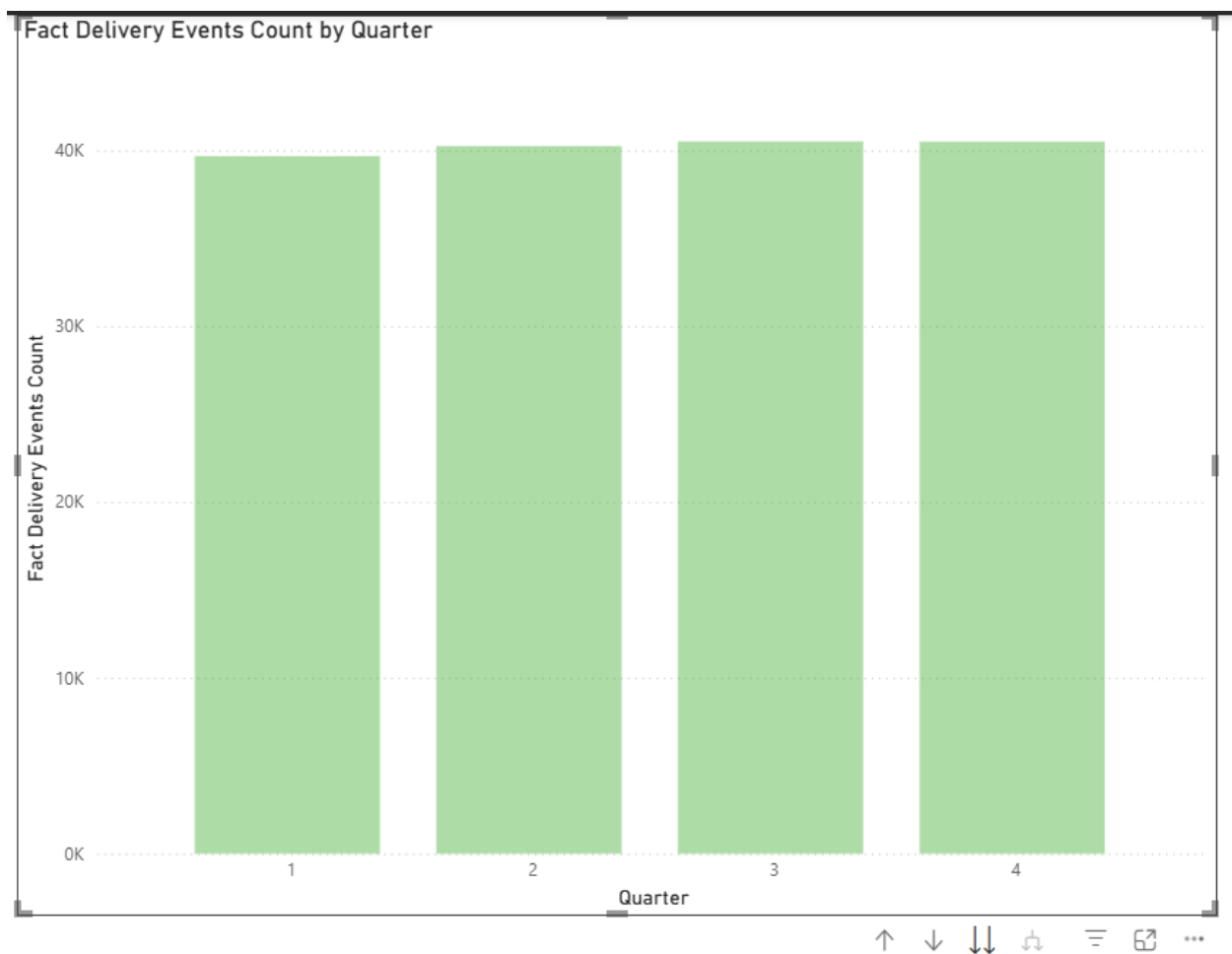


Figure 59 - Executing a drill-down to the second level of the hierarchy to analyze delivery volume distributions across Quarters.

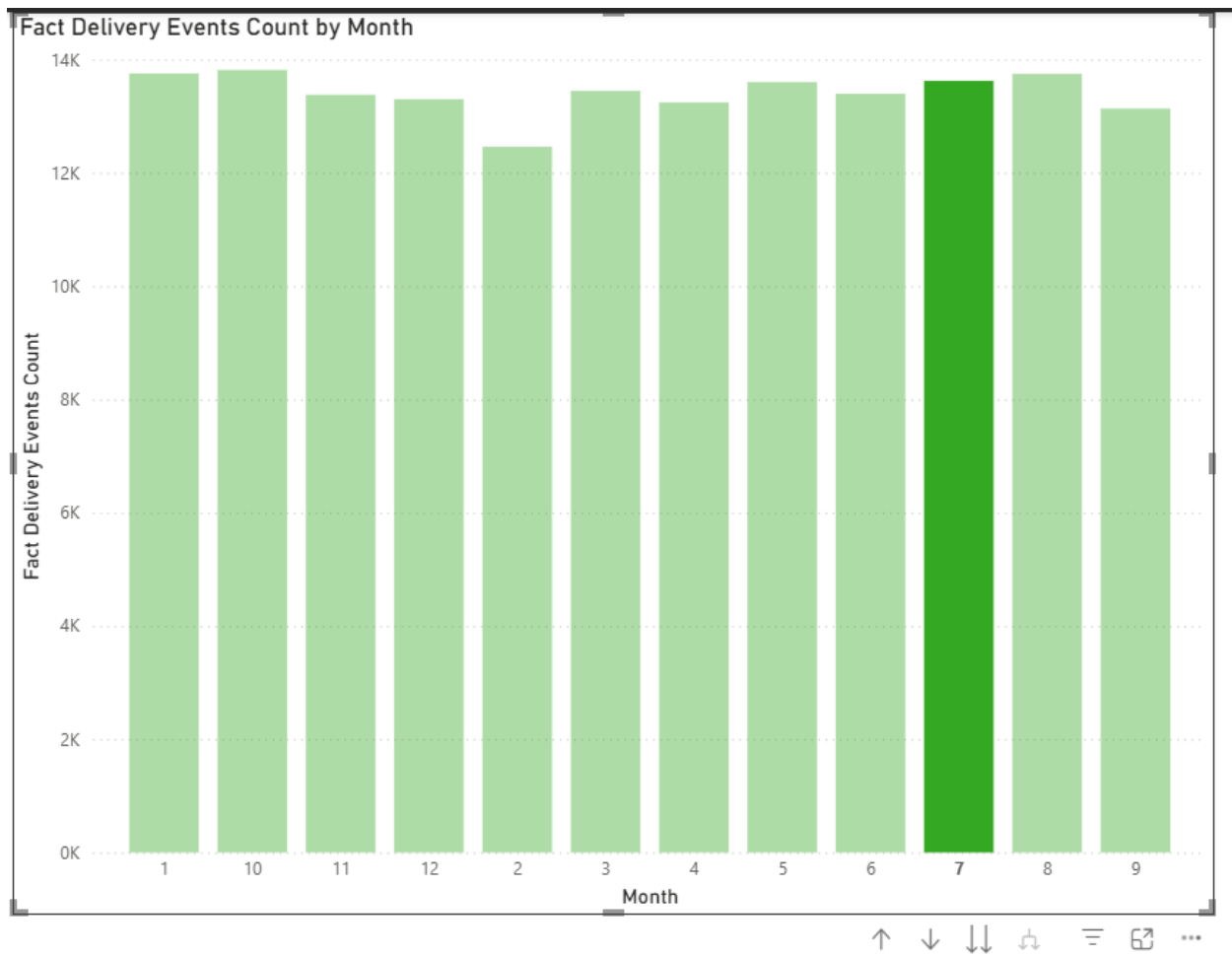


Figure 60 - Executing a further drill-down to the third level of the hierarchy, revealing more granular delivery trends across individual Months.

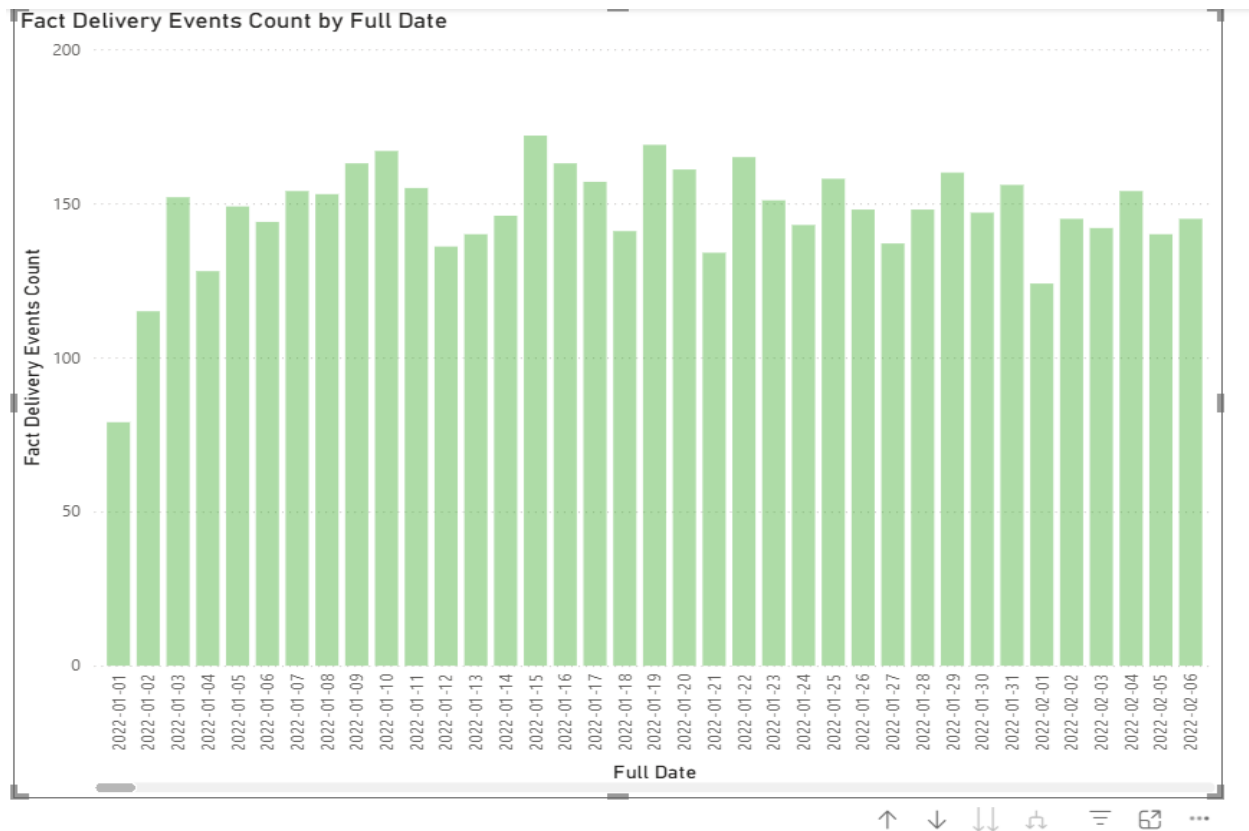


Figure 61 - Reaching the most granular level of the time hierarchy drill-down to view daily delivery event counts.

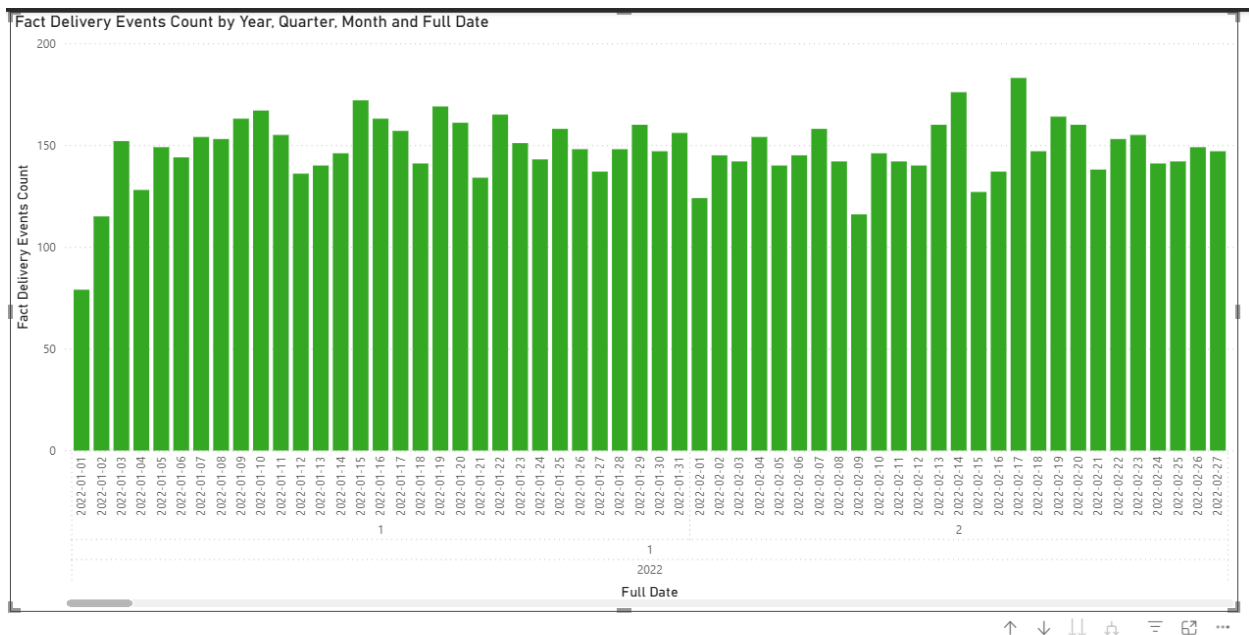


Figure 62 - Using the 'Expand all down' feature to display the complete concatenated time hierarchy (Year > Quarter > Month > Full Date) on a single continuous axis.

4. 6: Report 4: Drill-Through to Detailed Information

- **Objective:** Navigate from a high-level summary visual to a granular detail page.
- **Visual Design:** A multi-page setup utilizing a Summary Bar Chart and a Detailed Data Table.
- **Steps Followed:**
 1. Target Page Creation: Created a detail page containing a Table visual. Populated the table's Columns well with granular attributes: State Province, City, Year, and Fact Delivery Events Count.

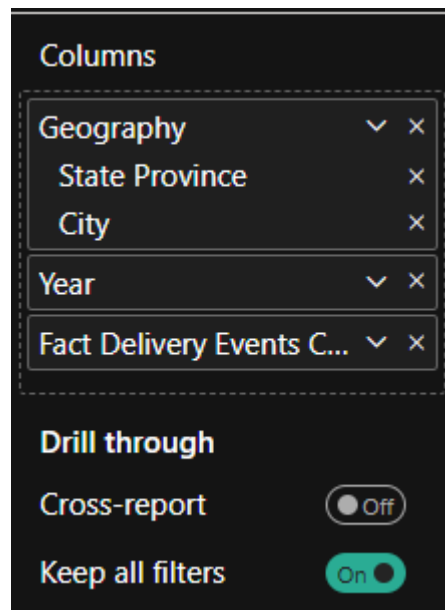


Figure 63 - Configuring the granular details Table visual on the target drill-through page by assigning spatial and temporal attributes to the Columns well.

State	Province	City	Year	Fact Delivery Events C
AZ		Phoenix	2022	2167
AZ		Phoenix	2023	2133
AZ		Phoenix	2024	2128
AZ		Phoenix	2025	5
CA		Los Angeles	2022	2189
CA		Los Angeles	2023	2188
CA		Los Angeles	2024	2177
CA		Los Angeles	2025	1
CO		Denver	2022	1064
CO		Denver	2023	1092
CO		Denver	2024	1086
CO		Denver	2025	1
FL		Miami	2022	3257
FL		Miami	2023	3183
FL		Miami	2024	3184
FL		Miami	2025	7
GA		Atlanta	2022	4246
GA		Atlanta	2023	4284
GA		Atlanta	2024	4212
GA		Atlanta	2025	5
IL		Chicago	2022	1038
IL		Chicago	2023	1042
IL		Chicago	2024	1103
IL		Chicago	2025	2
IN		Indianapolis	2022	4312
IN		Indianapolis	2023	4261
IN		Indianapolis	2024	4371
IN		Indianapolis	2025	4
MI		Detroit	2022	4345
MI		Detroit	2023	4164
MI		Detroit	2024	4358
MI		Detroit	2025	10
Total				160916

Figure 64 - The populated detail table displaying raw delivery event records across State, City, and Year prior to any drill-through filtering.

2. Drill-Through Setup: On the target page, dragged the State Province field into the Drill through section of the visualizations pane. This established the page as a destination and automatically generated a navigation back button.

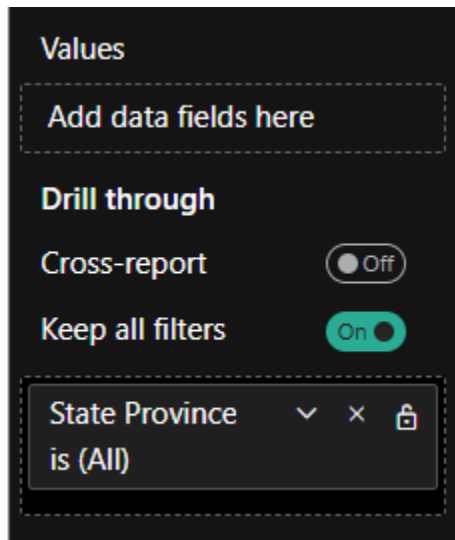
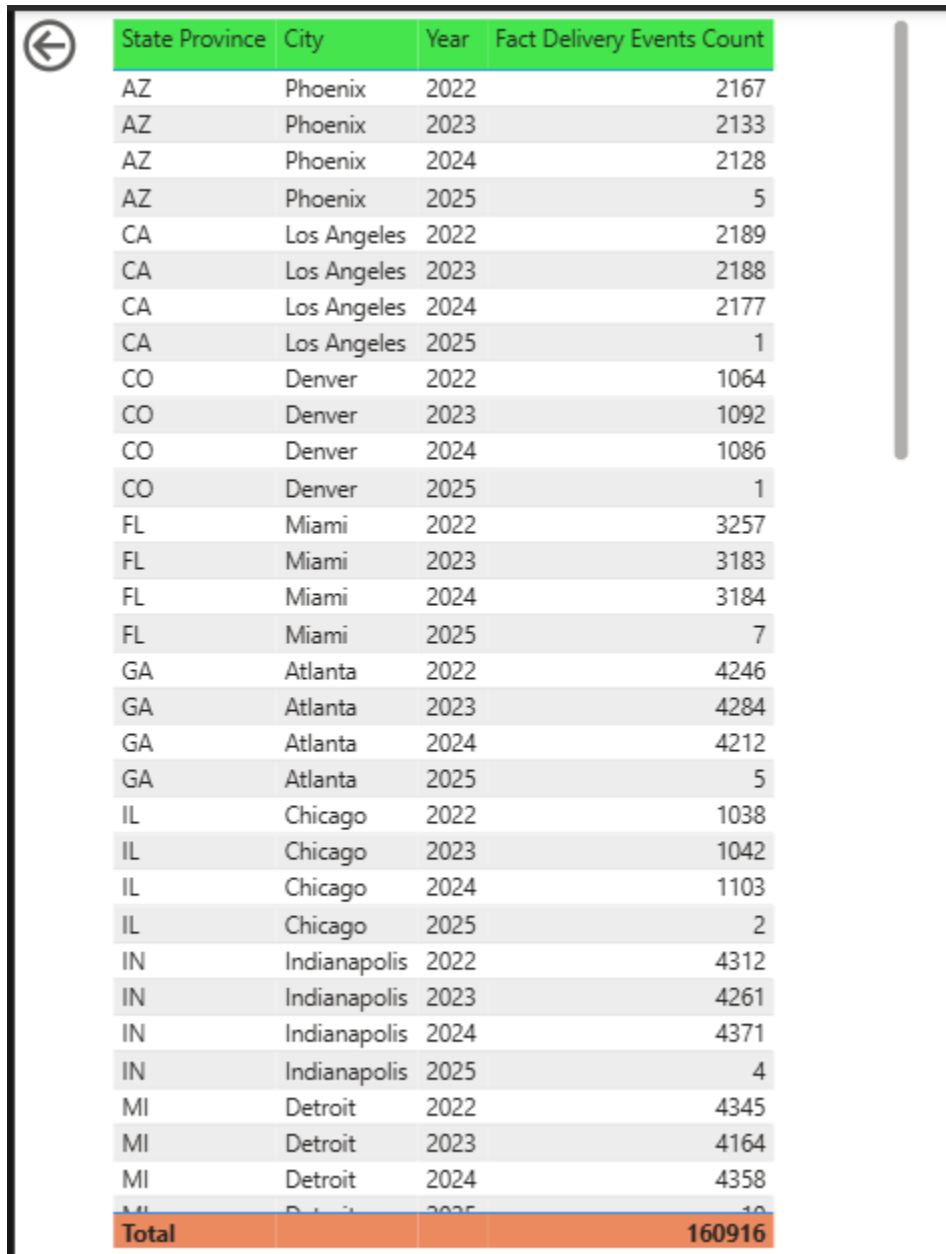


Figure 65 - Designating the page as a drill-through target by dragging the State Province field into the Drill-through section of the visualizations pane.



The screenshot shows a web application interface. On the left, there is a navigation menu with a back button (a circle containing a left-pointing arrow). The main content area displays a table with four columns: 'State Province', 'City', 'Year', and 'Fact Delivery Events Count'. The table lists data for various cities across different years, with a total row at the bottom. A vertical scrollbar is visible on the right side of the table.

State Province	City	Year	Fact Delivery Events Count
AZ	Phoenix	2022	2167
AZ	Phoenix	2023	2133
AZ	Phoenix	2024	2128
AZ	Phoenix	2025	5
CA	Los Angeles	2022	2189
CA	Los Angeles	2023	2188
CA	Los Angeles	2024	2177
CA	Los Angeles	2025	1
CO	Denver	2022	1064
CO	Denver	2023	1092
CO	Denver	2024	1086
CO	Denver	2025	1
FL	Miami	2022	3257
FL	Miami	2023	3183
FL	Miami	2024	3184
FL	Miami	2025	7
GA	Atlanta	2022	4246
GA	Atlanta	2023	4284
GA	Atlanta	2024	4212
GA	Atlanta	2025	5
IL	Chicago	2022	1038
IL	Chicago	2023	1042
IL	Chicago	2024	1103
IL	Chicago	2025	2
IN	Indianapolis	2022	4312
IN	Indianapolis	2023	4261
IN	Indianapolis	2024	4371
IN	Indianapolis	2025	4
MI	Detroit	2022	4345
MI	Detroit	2023	4164
MI	Detroit	2024	4358
MI	Detroit	2025	10
Total			160916

Figure 66 - The target detail page automatically generating a navigation Back button after the drill-through parameter is successfully configured.

3. Source Page Creation: Created a main summary page with a Clustered Bar Chart. Configured the visual by placing State Province on the Y-axis and Fact Delivery Events Count on the X-axis.

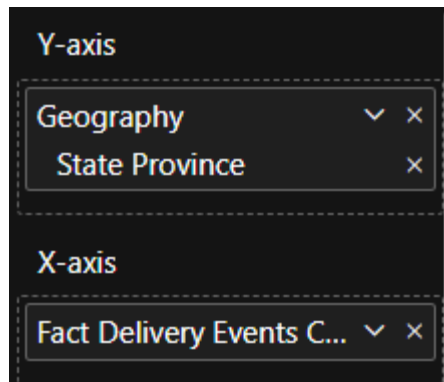


Figure 67 - Configuring the Source page Clustered Bar Chart to act as the primary navigation visual for the drill-through action.

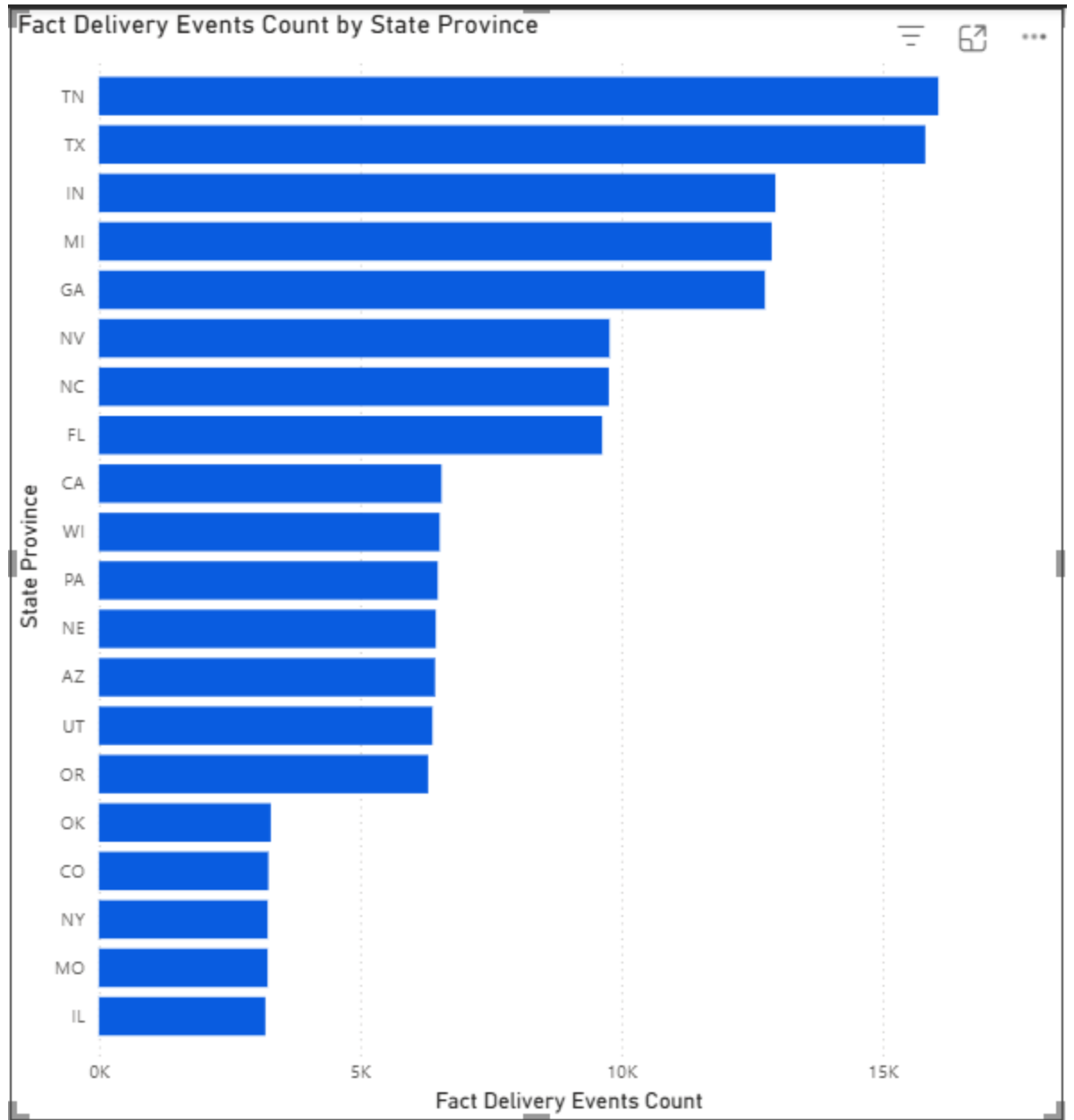


Figure 68 - The completed Source page Summary Bar Chart, displaying high-level delivery counts aggregated by State Province.

- **Outcome:** Users can right-click any specific State bar on the summary chart (e.g., Texas/TX) and select "Drill through" to automatically navigate to the target page. As demonstrated in the testing phase, the target table dynamically filters to display only the specific city and yearly data for the selected state.

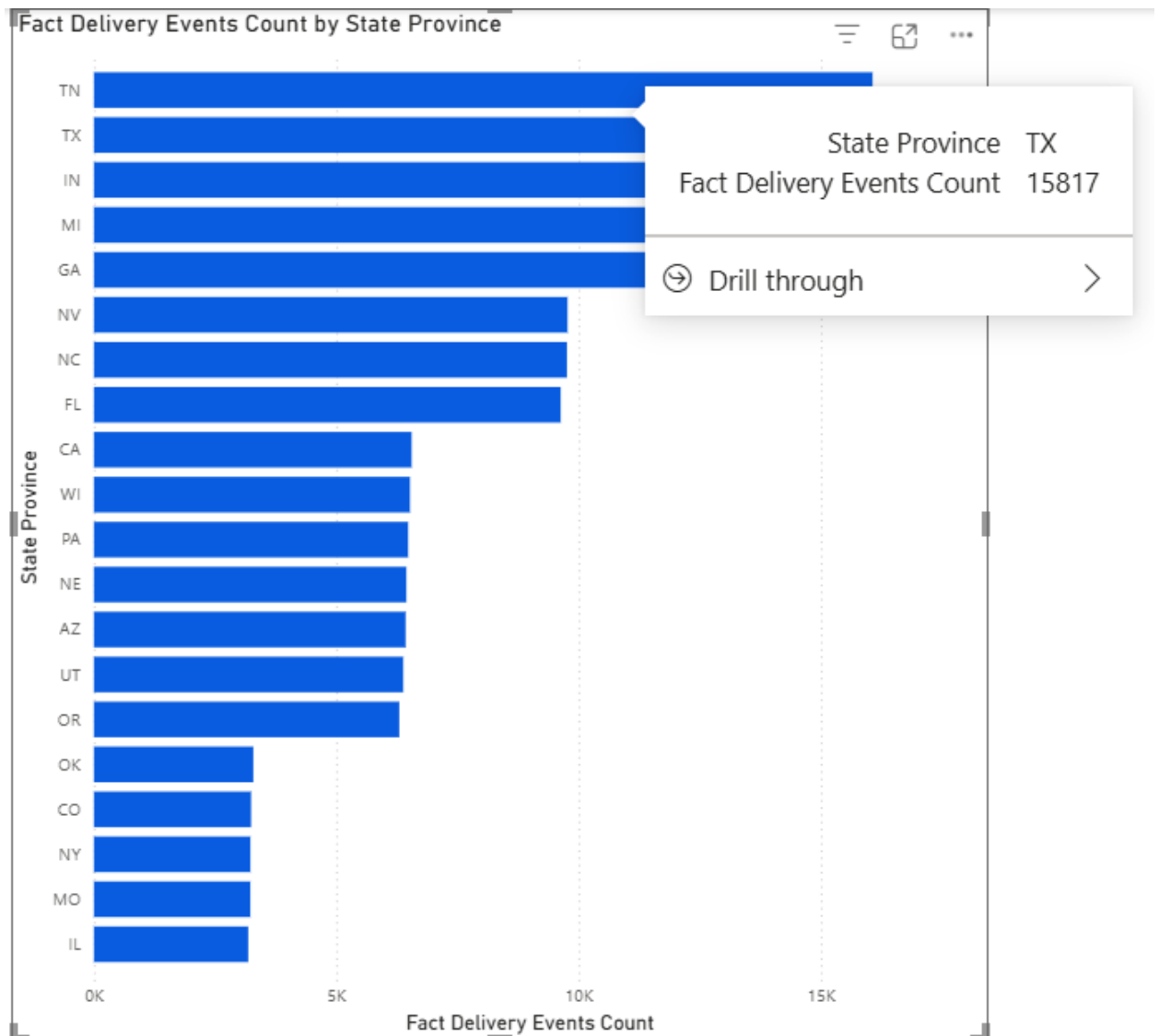


Figure 69 - Demonstrating the drill-through functionality by right-clicking the 'TX' (Texas) data point to navigate directly to its filtered granular records.